TRIANGULAR CENTROIDS OF EXPONENTIAL WALKS ON RIEMANN ZETA ORDINATES AND THE RIEMANN–VON MANGOLDT CLOCK

UNCONDITIONAL LINEARIZATION, A SMOOTH-CLOCK LIMIT, AND A CENTROID-FAMILY EQUIVALENCE TO RH

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ABSTRACT. Let $0 < \gamma_1 \le \gamma_2 \le \cdots$ be the positive ordinates of the nontrivial zeros of the Riemann zeta function and let

$$C_N^{\zeta} = \frac{1}{N} \sum_{k \le N} (N - k + 1) e^{i\gamma_k}. \tag{0.1}$$

The sequence γ_k enumerates every upper-half-plane zero individually with algebraic multiplicity; equal positive ordinates are therefore repeated. We separate this ordinal triangular centroid from the centroid of the synthetic Riemann–von Mangoldt clock. With the exact Riemann–Siegel counting clock $H(x) = 1 + \vartheta(x)/\pi$ and the midpoint argument term $S_*(\gamma) = \mathcal{N}_*(\gamma) - H(\gamma)$ at a distinct positive ordinate γ, an unconditional Stieltjes decomposition gives

$$C_N^{\zeta} - Q_N^{(H)} \longrightarrow 0. \tag{0.2}$$

The same reduction proves that the true-zero Landau-weighted statistic

$$L_N^{(M)} = \sum_{k \le N} \left(1 - \frac{M(\gamma_k)}{M(\gamma_N)}\right) e^{i\gamma_k} \tag{0.3}$$

has the same asymptotic target: $L_N^{(M)} - C_N^{\zeta} \to 0$ unconditionally. This does not replace γ_k by the Lambert inverse levels τ_k.

For the synthetic clock $M(\tau_k) = k - \frac{1}{2}$, the inverse is explicit through Lambert W; its exponential centroid is proved to converge and is represented by a conditionally convergent Abel series. For the true zeta centroid, the symmetric explicit formula at logarithmic frequency $u = 1$ yields, under RH and a stated Riesz-compatibility hypothesis,

$$\Re C_{\infty}^{\zeta} = \frac{1}{2\sqrt{e}} \left(e - \frac{1}{e^2 - 1}\right), \tag{0.4}$$

and a separate one-sided boundary-interchange hypothesis gives the conditional complex value

$$C_{\infty}^{\zeta} = 0.77689426079325\ldots + 0.76785566886455\ldots i. \tag{0.5}$$

The same ordinal centroid extends to the complex-frequency family $C_N^{\zeta}(u)$. For $\Im u > 0$ both the ordinate centroid and a completed-zero companion have absolutely convergent holomorphic limits. On the positive imaginary axis their difference is a positive Laplace sum, and its vanishing at any one fixed iy, $0 < y \le 4\pi$, is equivalent to RH. The criterion is therefore obtained from the centroid family itself. The experimentally observed point $u = 1$ is a distinct real-boundary statistic and is not identified with the interior completed-zero series.

Keywords. Riemann zeta function; zero ordinates; exponential sums; triangular centroid; Cesàro summation; Riesz means; Riemann–von Mangoldt formula; Lambert W; Riemann–Siegel theta function; explicit formula; Selberg second moment; Landau–Gonek formula; Fujii horizontal moment; Riemann hypothesis.

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List of notation

The recurring symbols are fixed as follows. Dummy variables used only inside a single integral or finite sum are not listed.

Symbol	Meaning
$k = 1, 2, \dots, N$	Ordinal zero index. Distinct indices may have the same ordinate.
N	Final value of the ordinal index k in a finite truncation; it is not a second running index.
$\rho_k = \beta_k + i\gamma_k$	Nontrivial zeros of ζ, with $\gamma_k > 0$, ordered by nondecreasing ordinate and counted with multiplicity.
$\nu(\gamma)$	Total upper-half-plane multiplicity at a positive ordinate γ; it is the jump size of $\mathcal{N}$ there.
$\delta_\rho = \beta - \frac{1}{2}$	Horizontal displacement of a nontrivial zero from the critical line. The symbol δ is reserved for this horizontal displacement throughout the analytic part of the paper.
$\mathcal{N}(x)$	Number of upper-half-plane nontrivial zeros with $0 < \gamma \le x$, each zero counted with algebraic multiplicity.
$\mathcal{N}_*(\gamma)$	Midpoint of the jump of $\mathcal{N}$ at a positive ordinate γ.
$\vartheta(x)$	Riemann–Siegel theta function.
$H(x) = 1 + \vartheta(x)/\pi$	Exact smooth Riemann–Siegel counting clock.
$S(x)$, $S_*(\gamma)$	Classical argument term in $\mathcal{N}(x) = H(x) + S(x)$ away from zero ordinates and its midpoint value at a zero ordinate γ.
r_k	Ordinal rank defect $r_k = k - \frac{1}{2} - H(\gamma_k)$; this is the quantity entering the exact finite centroid identity.
$r_k^{(M)}$	Main-clock ordinal defect $k - \frac{1}{2} - M(\gamma_k)$.
$M(x)$	Principal smooth Riemann–von Mangoldt clock $\frac{x}{2\pi}\log\frac{x}{2\pi e} + \frac{7}{8}$.
τ_k	Lambert inverse level defined by $M(\tau_k) = k - \frac{1}{2}$.
$\varepsilon_k = \gamma_k - \tau_k$	Vertical zero–RVM discrepancy; this symbol is kept distinct from $\delta_\rho = \beta - \frac{1}{2}$.
$b_k = e^{i\gamma_k} - e^{i\tau_k}$	Paired phase residual obtained by subtracting after exponentiation.
$a_k = e^{i\varepsilon_k} - 1$	Centered intrinsic residual obtained by subtracting the ordinates first and then exponentiating.
R_k^{RVM}, A_k^{RVM}	Running partial sums $\sum_{j\le k} b_j$ and $\sum_{j\le k} a_j$, $k = 1, \dots, N$.
$\mathcal{R}_{\mathrm{RVM}}(y)$, $\mathcal{A}_{\mathrm{RVM}}(y)$	Damped zero–RVM residual and its completed-zero comparison in Corollary 10.4.
$\widehat{\tau}_k$	Exact smooth level defined by $H(\widehat{\tau}_k) = k - \frac{1}{2}$.
g_k	Smooth-clock gap $g_k = \tau_{k+1} - \tau_k$.
v_k	Smooth-clock phase $v_k = e^{i\tau_k}$.
d_k, q_k	Abel coefficients for the synthetic Lambert walk: $d_k = (e^{ig_k} - 1)^{-1}$ and $q_k = d_{k+1} - d_k$.
S_N^ζ	Zeta exponential walk $\sum_{k\le N} e^{i\gamma_k}$.
C_N^ζ	Original triangular centroid at real frequency $u = 1$.
$w_{k,N}$	Triangular weight $1 - (k-1)/N$.
$Q_N^{(H)}$, $Q_N^{(M)}$	Linearized true-zero statistics using the exact clock H or the main clock M.
$L_N^{(M)}$	True-zero Landau-weighted statistic; the ordinates remain the actual γ_k.
M_N^{end}	Endpoint clock value $M(\gamma_N)$ used in the proof of the true-zero weighted target.

Symbol	Meaning
S_N^M, C_N^M	Exponential walk and triangular centroid of the synthetic Lambert levels τ_k.
$C_N^\zeta(u)$	Complex-frequency extension of the original ordinal centroid; $C_N^\zeta(1) = C_N^\zeta$.
$C^\zeta(u)$	Holomorphic interior limit of $C_N^\zeta(u)$ for $\Im u > 0$.
$C_N^\xi(u)$	Finite completed-zero centroid using the same triangular weights and the factors $e^{u(\rho_k - 1/2)}$.
$C^\xi(u)$	Holomorphic interior limit of $C_N^\xi(u)$ for $\Im u > 0$.
$\mathcal{D}(y)$	Positive imaginary-axis defect $C^\zeta(iy) - C^\xi(iy)$, $y > 0$.
$\mathcal{B}(x_0)$, $\mathcal{P}_{\infty,\sigma}(x_0)$	Smooth and arithmetic pieces of the one-sided boundary transform, with auxiliary height cutoff $x_0 < \gamma_1$.
C_∞^ζ	Conditional ordinal-centroid limit under RH and the stated boundary-interchange hypothesis; it is distinct from the interior series $C^\xi(u)$.
$A_2(x)$	Fujii horizontal second-moment counting function with auxiliary height cutoff x.
$H_2(s)$, $F_2(z)$	Dirichlet and Laplace transforms of the horizontal second-moment mass.
RH	Riemann hypothesis.

1. Introduction

The present work grew from a simple numerical observation. I considered the exponential walk obtained by placing the positive ordinates of the nontrivial zeros of the Riemann zeta function on the unit circle and successively adding the corresponding vectors, and then compared its trajectory centroid with the centroid obtained when the same ordinal indices are replaced by a Lambert inversion of the smooth Riemann–von Mangoldt clock. The two constructions use almost indistinguishable ordinates at large height, but their centroids are strikingly different. This was the starting point of the paper: to understand what produces that difference, what part comes from the smooth counting law, what part belongs to the actual zeros, and how the centroid changes when the true-zero sequence is replaced by its smooth ordinal counterpart.

Write

$$0 < \gamma_1 \le \gamma_2 \le \cdots$$

for the positive ordinates, counted with multiplicity, and let τ_k denote the paired Lambert–RVM levels defined by

$$M(\tau_k) = k - \frac{1}{2}, \qquad M(x) = \frac{x}{2\pi}\log\frac{x}{2\pi e} + \frac{7}{8}. \tag{1.1}$$

At frequency one define

$$S_k^\zeta = \sum_{j=1}^{k} e^{i\gamma_j}, \qquad S_k^{\mathrm{RVM}} = \sum_{j=1}^{k} e^{i\tau_j}, \qquad k = 1, \dots, N, \tag{1.2}$$

and their triangular trajectory centroids by

$$C_N^\zeta = \frac{1}{N}\sum_{k=1}^{N} S_k^\zeta, \qquad C_N^{\mathrm{Lamb}} = \frac{1}{N}\sum_{k=1}^{N} S_k^{\mathrm{RVM}}. \tag{1.3}$$

The numerical work uses the first 2,001,052 positive zeta-zero ordinates in Odlyzko's public table, together with the paired Lambert sequence on exactly the same ordinal range [15]. For this full

dataset,

$$C_N^{\zeta} = 0.776907119554533 + 0.767844590760585\, i,$$
$$C_N^{\mathrm{Lamb}} = 0.041570219277355 + 1.757541749592341\, i. \tag{1.4}$$

Thus

$$C_N^{\zeta} - C_N^{\mathrm{Lamb}} = 0.735336900277178 - 0.989697158831756\, i. \tag{1.5}$$

The same computation exposes two different residual constructions, which must not be confused. Subtracting after exponentiation gives

$$b_k := e^{i\gamma_k} - e^{i\tau_k}, \qquad R_k^{\mathrm{RVM}} := \sum_{j\le k} b_j, \tag{1.6}$$

whereas subtracting the ordinates first and then exponentiating gives

$$\varepsilon_k := \gamma_k - \tau_k, \qquad U_k^{\mathrm{RVM}} := \sum_{j\le k} e^{i\varepsilon_j}. \tag{1.7}$$

Its centered intrinsic residual is

$$a_k := e^{i\varepsilon_k} - 1, \qquad A_k^{\mathrm{RVM}} := \sum_{j\le k} a_j. \tag{1.8}$$

The exact factorization

$$\boxed{b_k = e^{i\tau_k}\left(e^{i\varepsilon_k} - 1\right) = 2i\, e^{i(\gamma_k+\tau_k)/2} \sin\frac{\varepsilon_k}{2}} \tag{1.9}$$

shows that $|b_k| = 2|\sin(\varepsilon_k/2)|$: its modulus sees only the wrapped vertical discrepancy, while the orientation is imposed by the RVM carrier $e^{i\tau_k}$.

For the same full dataset,

$$R_N^{\mathrm{RVM}} = 0.774915676816428 - 2.275776686742230\, i, \qquad \max_{k\le N} |R_k^{\mathrm{RVM}}| = 3.006826277026504, \tag{1.10}$$

while

$$\left(\sum_{k\le N} |b_k|^2\right)^{1/2} = 264.9282445682294. \tag{1.11}$$

The corresponding triangular residual centroid is

$$\boxed{\frac{1}{N}\sum_{k\le N} R_k^{\mathrm{RVM}} = C_N^{\zeta} - C_N^{\mathrm{Lamb}} = 0.735336900277166 - 0.989697158831749\, i,} \tag{1.12}$$

and the triangular centroid of the centered intrinsic residual is

$$\boxed{\frac{1}{N}\sum_{k\le N} A_k^{\mathrm{RVM}} = -18783.994978046514007 - 0.175254737451056\, i.} \tag{1.13}$$

These numerical objects are displayed below through four trajectory figures. In each case the three-dimensional ordered walk is accompanied by its projection onto the complex plane, drawn with very thin lines so that the geometry of the path remains visible. All four figures are generated directly from the same 2,001,052 paired binary ordinates by the Python program reproduced in Appendix L; its complete console output is reproduced in Appendix M.

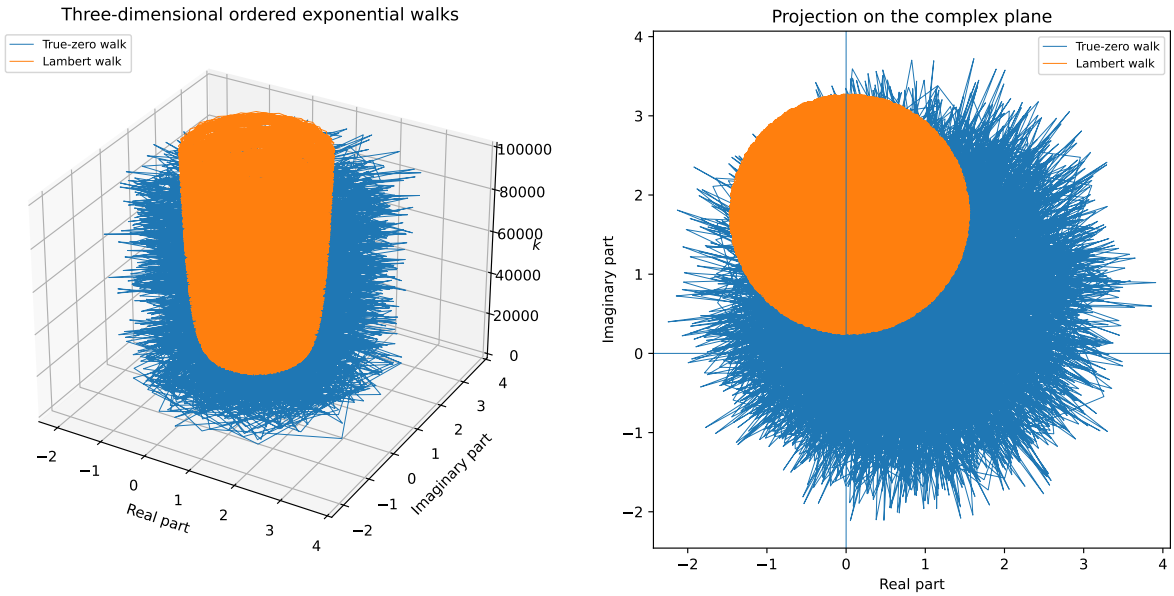


Figure 1. Ordered exponential walks for the first 100,000 ordinates. The left panel is the three-dimensional trajectory and the right panel is its projection onto the complex plane. The two curves are the true-zero walk S_k^{ζ} and the paired Lambert walk S_k^{RVM}, shown on the same complex scale; the vertical coordinate in the three-dimensional panel is the ordinal index k.

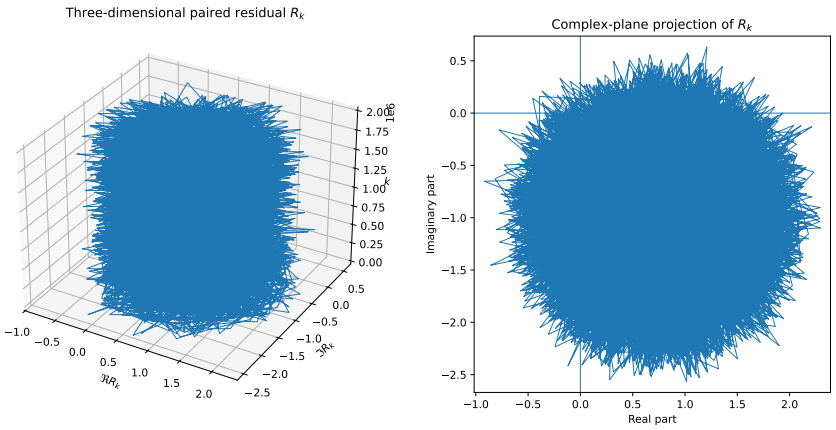


Figure 2. Paired residual obtained by subtracting after exponentiation: $R_k^{\mathrm{RVM}} = \sum_{j \le k} (e^{i\gamma_j} - e^{i\tau_j})$. The left panel is the three-dimensional trajectory and the right panel is its projection onto the complex plane. Over all 2,001,052 pairs the observed path remains inside $|R_k^{\mathrm{RVM}}| \le 3.0068262771$.

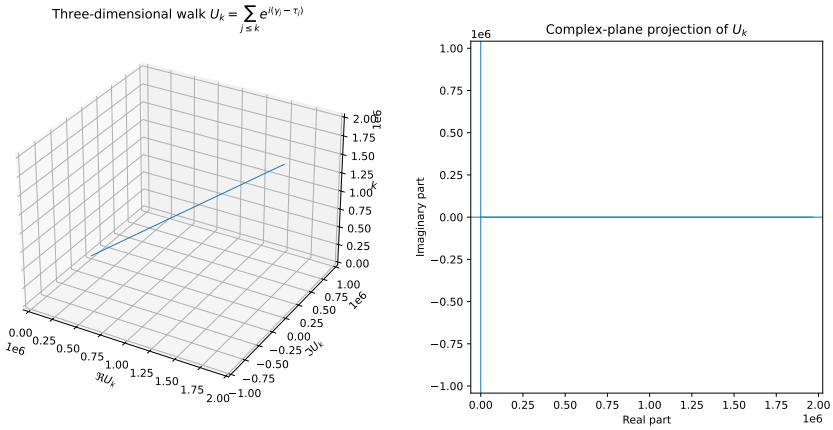


Figure 3. Difference before exponentiation: $U_k^{\mathrm{RVM}} = \sum_{j \le k} e^{i(\gamma_j - \tau_j)}$. The left panel is the three-dimensional trajectory and the right panel is its projection onto the complex plane. This is not the same object as Figure 2; its centered version is $A_k^{\mathrm{RVM}} = U_k^{\mathrm{RVM}} - k$.

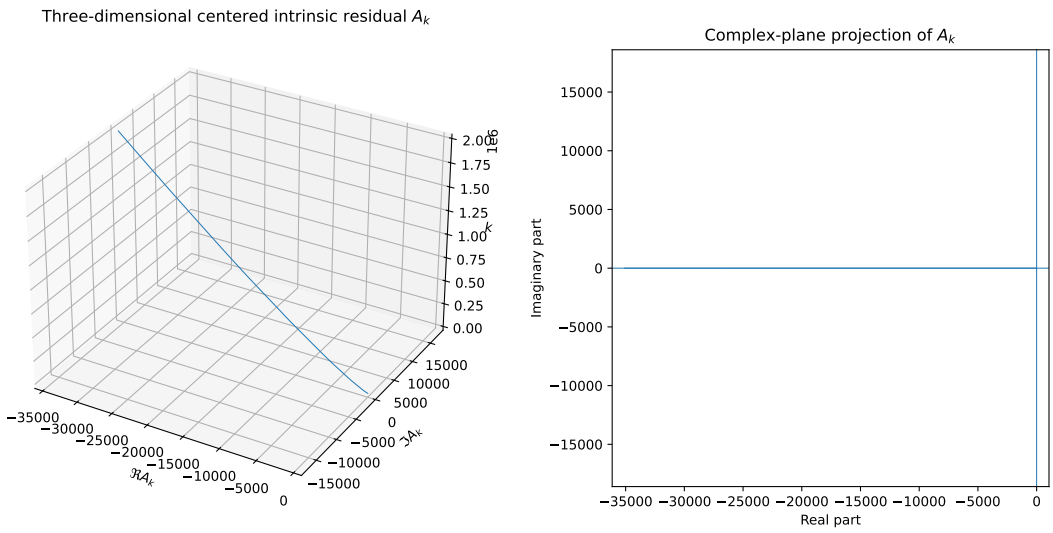


Figure 4. Centered intrinsic residual: $A_k^{\mathrm{RVM}} = \sum_{j\le k}(e^{i(\gamma_j-\tau_j)}-1)$. The left panel is the three-dimensional trajectory and the right panel is its projection onto the complex plane.

The centroid is the first Cesàro mean of the endpoint sequence and, equivalently, a Riesz mean of order one relative to the ordinal frequency $\lambda_k = k-1$. The triangular factor is not a fixed cutoff that protects a finite prefix: for each fixed k, $1-(k-1)/N \to 1$. Consequently, stabilization of C_N^{ζ} is a cancellation statement involving the whole expanding spectrum.

The frequency itself is special. The phase $e^{i\gamma}$ is the continuous Landau frequency $x = e$, because $x^{i\gamma} = e^{i\gamma}$ when $x = e$. Since e is not an integral prime power, the leading fixed-x prime-power resonance vanishes. Hlawka's equidistribution theorem gives $N^{-1}S_N^{\zeta} \to 0$, while Ford–Zaharescu and Ford–Soundararajan–Zaharescu provide refined information on fixed multiples of zero ordinates and non-resonant secondary terms [8, 11, 12]. What those results do not directly evaluate is the order-one constant selected by the moving ordinal Cesàro kernel.

A distinction used throughout the paper is essential. Weighting the *true* ordinates by the smooth counting clock, for example through

$$\sum_{k=1}^{N}\left(1-\frac{M(\gamma_k)}{M(\gamma_N)}\right)e^{i\gamma_k}, \tag{1.14}$$

does not replace the ordinates by the Lambert levels τ_k. Theorem 5.3 below shows that the statistic in (1.14) has the same asymptotic target as C_N^{ζ}. Its numerical benchmark is therefore the zeta value $0.776907119554635 + 0.767844590760710\,i$, not the synthetic value C_N^{M}.

There are three layers to the answer.

First, the ordinal triangular centroid is superficially nonlinear in the zero-counting measure because the coefficient $N-k+1$ contains the rank k. The index is literal: if a positive ordinate γ occurs with total upper-half-plane multiplicity $\nu(\gamma)$, then it occupies $\nu(\gamma)$ consecutive positions and every later index is shifted accordingly. On RH this includes ordinary multiplicity on the critical line; off RH it also includes the two upper-half-plane members $\beta+i\gamma$ and $1-\beta+i\gamma$ of each functional-equation pair, with their algebraic multiplicities. The lower conjugates at $-\gamma$ are not indexed. We therefore distinguish the midpoint argument term $S_*(\gamma) = \mathcal{N}_*(\gamma) - H(\gamma)$ from the individual ordinal defects

$$r_k := k - \frac{1}{2} - H(\gamma_k).$$

With

$$H(x) = 1 + \frac{\vartheta(x)}{\pi},$$

the exact decomposition is

$$C_N^{\zeta} = Q_N^{(H)} - \frac{1}{N} \sum_{k \le N} r_k e^{i\gamma_k}.$$

Over a complete ordinate block, the ordinal defects sum to $\nu(\gamma)$ times the midpoint argument value. This makes the Stieltjes analysis exact without assuming simplicity. We prove unconditionally that the normalized second term tends to zero. Thus the asymptotic centroid is reduced, without RH, to a linear statistic of the ordinates. This is the "Level 1" reduction in the terminology of the computational development.

Second, the synthetic smooth clock can be solved completely. For the main Riemann–von Mangoldt approximation

$$M(x) = \frac{x}{2\pi} \log \frac{x}{2\pi e} + \frac{7}{8},$$

the inverse ordinates are explicit through Lambert W. Their gaps are slowly decreasing and the associated phases admit a discrete Abel identity. Combining the first-derivative exponential-sum estimate with a second summation by parts proves convergence of the RVM centroid and yields a rapidly converging series for its limit. This is an unconditional theorem about the smooth model, independent of RH.

Third, evaluating the true zeta centroid by a holomorphic explicit formula introduces a different issue. An ordinary explicit formula naturally sees a zero $\rho = \beta + i\gamma$ through a holomorphic weight such as $e^{\rho - 1/2}$, whereas the centroid uses the pure ordinate phase $e^{i\gamma}$. These coincide on the critical line. Off the line, functional-equation pairing produces the factor

$$\cosh\left(\beta - \frac{1}{2}\right).$$

This is not an artefact of a particular contour; $e^{i \operatorname{Im} s}$ is not holomorphic. In particular, there cannot be a universal holomorphic kernel on the critical strip whose imaginary part is $\sin t$ independently of the real coordinate, because $\sin t$ is not harmonic as a function of (σ, t).

This observation determines the precise role of RH. RH is not required to define C_N, nor to prove the unconditional Level 1 reduction. RH is required if one wants to replace the ordinate-only phase by the standard holomorphic complete-zero phase without retaining a horizontal defect. Under RH, a symmetric explicit-formula calculation at $x = e$ gives the closed real coordinate

$$\frac{1}{2\sqrt{e}} \left(e - \frac{1}{e^2 - 1} \right).$$

The underlying regularized bilateral identity is classical; the new point is its interpretation as the real coordinate of this particular ordinal triangular centroid, subject to the exact summability compatibility stated below.

The imaginary coordinate is one-sided. Pairing $+\gamma$ and $-\gamma$ removes it identically, so it cannot be recovered from the symmetric explicit formula alone. A one-sided transform of $S(t)$ produces instead a boundary value of $\operatorname{Log} \zeta$ with the two frequency branches $1 + \log n$ and $1 - \log n$. This is closely related in spirit to generalized Littlewood formulae and to the logarithmic-integral criteria of Sekatskii–Beltraminelli–Merlini [27, 28, 29], but the present observable is an ordinal Cesàro transform rather than a fixed analytic test function. The exact bridge between those summation procedures is therefore stated as a separate boundary-interchange hypothesis rather than hidden inside the proof.

Finally, the horizontal defect connects the problem to a well-developed line of work. Fujii related antiderivatives of $S(t)$ to the positive quantity

$$A_2(x) = \sum_{\substack{\beta>1/2 \\ 0<\gamma\le x}} \left(\beta - \frac{1}{2}\right)^2,$$

and Bondarenko–Ivić–Saksman–Seip showed that this same horizontal mass governs the analytic continuation of the spectral zeta function

$$G(s) = \sum_{\gamma>0} \gamma^{-s}$$

beyond its unconditional half-plane [20, 22]. The centroid defect begins with exactly this quadratic mass, but with an oscillatory factor $e^{i\gamma}$. At the boundary frequency $u = 1$, zero-density estimates control its absolute mass but not the corresponding oscillatory value.

The organization is as follows. Sections 2–3 set the counting and summability conventions. Section 4 proves the unconditional linearization. Section 5 compares the exact Riemann–Siegel clock with its Lambert approximation. Section 6 gives the unconditional RVM centroid theorem. Sections 7–8 derive the symmetric and one-sided explicit-formula values under precisely stated hypotheses. Section 9 isolates the off-critical-line defect. Section 10 promotes the same centroid to the holomorphic family $C_N^{\zeta}(u)$ and proves the imaginary-axis equivalence with RH. Section 11 relates the horizontal defect to Fujii's moment and the spectral zeta function. The final numerical section records the finite values, figures, and reproducibility checks.

2. Preliminaries and conventions

2.1. **Zeros, ordinal enumeration, multiplicity blocks, and the counting function.** Throughout the paper the ordinal index is

$$\boxed{k = 1, 2, 3, \dots, N,}$$

where k is the current zero index and N is only the final value of that same index in a finite truncation. We never use N as a second running index. Let

$$\rho_k = \beta_k + i\gamma_k, \qquad \gamma_k > 0,$$

enumerate *all* nontrivial zeros in the upper half-plane, counted with their individual algebraic multiplicities and ordered by nondecreasing ordinate. Thus

$$0 < \gamma_1 \le \gamma_2 \le \cdots \le \gamma_N,$$

and γ_1 is literally the ordinate of the first upper-half-plane zero. The indices themselves are always distinct: there are never two equal values of k. What can be equal are their ordinates, for example $\gamma_k = \gamma_{k+1}$. Every repeated ordinate occupies a separate ordinal slot and therefore shifts every subsequent index. The conjugate zeros with ordinate $-\gamma_k$ are not included in this enumeration.

For any positive zero ordinate γ, define its *upper-half-plane ordinate multiplicity*

$$\nu(\gamma) := \sum_{\substack{0<\beta<1 \\ \zeta(\beta+i\gamma)=0}} \operatorname{mult}(\beta + i\gamma). \tag{2.1}$$

If

$$a(\gamma) := \#\{k : \gamma_k < \gamma\}, \qquad b(\gamma) := a(\gamma) + \nu(\gamma),$$

then the whole ordinate block is

$$\gamma_{a(\gamma)+1} = \cdots = \gamma_{b(\gamma)} = \gamma. \tag{2.2}$$

This convention treats multiplicity on and off the critical line without changing the indexing rule. On RH, a zero $1/2 + i\gamma$ of multiplicity m contributes m copies of the same ordinate γ. Off the

critical line, a zero $\beta + i\gamma$, $\beta \neq 1/2$, of multiplicity m is accompanied by $1 - \beta + i\gamma$ with the same multiplicity, so the generic functional-equation pair contributes $2m$ copies of the *same positive ordinate* γ. Their lower-half-plane conjugates contribute no additional indices here. Consequently every later ordinal index is shifted by the full upper-half-plane multiplicity actually present at preceding ordinates. The paired smooth level τ_k is always attached to this actual ordinal index k.

This is exactly where a conversion from a classical height cutoff to the ordinal cutoff must be handled with care. If a common ordinate occupies the consecutive slots

$$\gamma_k = \gamma_{k+1} = \cdots = \gamma_{k+m-1} = \gamma,$$

and a finite ordinal truncation ends inside that block, then a height cutoff at γ includes the *entire* block, whereas the ordinal truncation includes only the slots up to N. Therefore a height sum at γ_N is not identified term by term with a sum through $k = N$ unless N is the last slot of its multiplicity block. The difference is precisely the uncompleted part of that one block. On the critical line a multiplicity-m zero creates m equal ordinates. Off the critical line a generic functional-equation pair of multiplicity m on each side creates $2m$ equal positive ordinates. In both cases those repetitions are part of the ordinal numbering itself, not a secondary correction to it.

Let

$$\mathcal{N}(x) := \#\{k : \gamma_k \leq x\}, \tag{2.3}$$

so every zero is counted individually with algebraic multiplicity. At a zero ordinate γ, let

$$\mathcal{N}_*(\gamma) := \frac{1}{2}(\mathcal{N}(\gamma^-) + \mathcal{N}(\gamma^+)) = a(\gamma) + \frac{\nu(\gamma)}{2}. \tag{2.4}$$

Define the Riemann–Siegel theta function

$$\vartheta(x) = \operatorname{Im} \operatorname{Log} \Gamma\left(\frac{1}{4} + \frac{ix}{2}\right) - \frac{x}{2} \log \pi,$$

with its standard continuous determination for $x > 0$, and put

$$H(x) = 1 + \frac{\vartheta(x)}{\pi}.$$

Away from zero ordinates the Riemann–von Mangoldt formula is

$$\mathcal{N}(x) = H(x) + S(x). \tag{2.5}$$

At an ordinate γ, define the midpoint argument value

$$S_*(\gamma) := \mathcal{N}_*(\gamma) - H(\gamma). \tag{2.6}$$

For every index $a(\gamma) + j$ in the block, $1 \leq j \leq \nu(\gamma)$, put

$$r_{a(\gamma)+j} := a(\gamma) + j - \frac{1}{2} - H(\gamma). \tag{2.7}$$

Then the complete-block identity is exact:

$$\boxed{\sum_{j=1}^{\nu(\gamma)} r_{a(\gamma)+j} = \nu(\gamma) S_*(\gamma).} \tag{2.8}$$

Thus Stieltjes jumps and ordinal weights agree on every complete ordinate block, whether that block comes from a multiple critical-line zero, an off-line functional-equation pair, or any combination of zeros sharing the same ordinate.

Lemma 2.1 (Height-to-index transport with multiplicity)**.** *Uniformly for indices $k \to \infty$,*

$$k = M(\gamma_k) + O(\log(2+\gamma_k)), \tag{2.9}$$

and therefore

$$k \asymp \gamma_k \log(2+\gamma_k), \qquad \gamma_k \asymp \frac{k}{\log(2+k)}. \tag{2.10}$$

The estimates remain valid when k lies inside a multiplicity block.

Proof. For the block containing γ_k, the index k lies between $a(\gamma_k)+1$ and $a(\gamma_k)+\nu(\gamma_k)$, hence

$$|k - \mathcal{N}_*(\gamma_k)| \le \frac{\nu(\gamma_k)+1}{2}.$$

The jump of $\mathcal{N}$ at γ_k equals $\nu(\gamma_k)$. Since H is continuous and the classical bound $S(x) = O(\log(2+x))$ holds on both sides of the jump,

$$\nu(\gamma_k) = O(\log(2+\gamma_k)), \qquad S_*(\gamma_k) = O(\log(2+\gamma_k)). \tag{2.11}$$

Thus $\mathcal{N}_*(\gamma_k) = H(\gamma_k) + O(\log(2+\gamma_k))$. Using $H(\gamma_k) = M(\gamma_k) + O(1/\gamma_k)$ gives (2.9). The standard asymptotic form of M then gives (2.10). No simplicity or RH assumption is used. □

Remark 2.2 (How external height estimates are used). The primary ordinal variable is $k = 1, 2, \dots, N$, with N used only for the final value of k. Classical theorems stated with a continuous height cutoff are auxiliary inputs and are never converted by the formal substitution "height = N". For a finite endpoint N, the relevant height is γ_N, but if N lies inside a repeated-ordinate block then the classical cutoff at γ_N includes the full block while the ordinal sum does not. We therefore (1) apply the external theorem to the complete block at γ_N; (2) remove the unused slots of that single endpoint block, counting their actual multiplicities; and only then (3) convert the height scale γ_N to the ordinal scale N by Lemma 2.1. This protocol is used explicitly below for Hlawka/Ford–Zaharescu, Selberg, the Landau–Gonek formula, and Fujii's horizontal moment, and it is valid both for multiplicity on the critical line and for repeated positive ordinates produced by off-line functional-equation pairs.

Proposition 2.3 (Ordinal forms of the height estimates used below)**.** *With the enumeration and multiplicity convention above, the external height estimates used in this paper imply the following ordinal statements.*

(i) Hlawka's equidistribution theorem gives

$$\sum_{k \le N} e^{i\gamma_k} = o(N).$$

(ii) At the non-resonant frequency $e^{i\gamma}$, the second-order Ford–Zaharescu theorem strengthens this to

$$\sum_{k \le N} e^{i\gamma_k} = o\left(\frac{N}{\log N}\right). \tag{2.12}$$

(iii) Selberg's height estimate

$$\int_0^x S(t)^2 \, \mathrm{d}t \ll x \log\log x$$

becomes, after evaluating at the endpoint ordinate and normalizing by N,

$$\frac{\gamma_N \log\log(2+\gamma_N)}{N} = O\left(\frac{\log\log N}{\log N}\right) = o(1).$$

(iv) Any complete-block error $O(\log^A(2+x))$, with fixed A, remains $o(N)$ after passing to the possibly partial block containing γ_N. In particular this applies to the local block corrections and to the $H - M$ clock replacement used below.

All sums in (i)–(iv) count the actual upper-half-plane zeros individually. Thus a critical-line zero of multiplicity m contributes m identical phases, while an off-line pair $\beta + i\gamma$, $1-\beta+i\gamma$, each of multiplicity m, contributes $2m$ identical ordinate phases $e^{i\gamma}$. These repetitions are part of N; they are not an error term and are never discarded in the height-to-index conversion.

Proof. For (i), Hlawka gives ordinary equidistribution of the fractional parts of $\gamma/(2\pi)$, with zeros counted with multiplicity. Applied at the complete height cutoff γ_N, the zero-mean test function $e^{2\pi iu}$ gives a complete-block sum $o(\mathcal{N}(\gamma_N))$. If N lies inside the terminal repeated-ordinate block, deleting the unused copies changes the sum by at most $\nu(\gamma_N) = O(\log(2+\gamma_N)) = o(N)$; hence the ordinal statement follows.

For (ii) we use precisely Corollary 2 of Ford–Zaharescu [11]. Their second-order formula, valid for every $f \in C^2(\mathbb{T})$, is

$$\sum_{0<\gamma\le\gamma_N} f(\alpha\gamma) = \mathcal{N}(\gamma_N)\int_{\mathbb{T}} f + \gamma_N \int_{\mathbb{T}} f(u) g_\alpha(u)\,\mathrm{d}u + o(\gamma_N),$$

where the sum is the complete height sum and zeros are counted with multiplicity. Take

$$f(u) = e^{2\pi iu}, \qquad \alpha = \frac{1}{2\pi}.$$

Then $\int_{\mathbb{T}} f = 0$. Moreover $g_\alpha \equiv 0$, because Ford–Zaharescu's resonant case requires $\alpha = (a/q)\log p/(2\pi)$, whereas $1 = (a/q)\log p$ would imply $e = p^{a/q}$, impossible. Thus

$$\sum_{0<\gamma\le\gamma_N} e^{i\gamma} = o(\gamma_N). \tag{2.13}$$

This sum contains the *whole* ordinate block at γ_N. The ordinal sum $\sum_{k=1}^{N} e^{i\gamma_k}$ may contain only part of that block; the difference has at most $\nu(\gamma_N) = O(\log(2+\gamma_N))$ unit-modulus terms. Hence it is still $o(\gamma_N)$. Finally Lemma 2.1 gives $\gamma_N \asymp N/\log N$, proving (2.12). This argument is unchanged whether the repeated endpoint comes from a critical-line zero of multiplicity m or from an off-line pair, which contributes the corresponding repeated positive-ordinate slots.

Parts (iii) and (iv) follow directly from Lemma 2.1 and (2.11), with the same terminal-block correction when the ordinal cutoff ends inside a repeated ordinate. □

2.2. **The main smooth clock and Lambert inversion.** We shall also use

$$M(x) = \frac{x}{2\pi}\log\frac{x}{2\pi e} + \frac{7}{8}. \tag{2.14}$$

The classical asymptotic expansion of ϑ gives

$$H(x) = M(x) + \frac{1}{48\pi x} + O(x^{-3}). \tag{2.15}$$

More terms and rigorous remainder estimates follow from the Stirling expansion for $\log\Gamma$; see, for example, Brent and Paris [30, 31].

If $M(\tau_k) = k - \frac{1}{2}$, then solving explicitly gives

$$\boxed{\tau_k = \frac{2\pi(k-11/8)}{W_0((k-11/8)/e)}}. \tag{2.16}$$

We call $\{\tau_k\}$ the Lambert RVM clock. The exact smooth clock $\{\widehat{\tau}_k\}$ is defined instead by

$$H(\widehat{\tau}_k) = k - \frac{1}{2}.$$

The two clocks are asymptotically close, but they are conceptually distinct.

3. The exponential walk and its triangular Riesz mean

Definition 3.1 (Zeta exponential walk and centroid)**.** For the running ordinal index $k = 1, 2, \dots, N$, define

$$S_k^\zeta := \sum_{j=1}^{k} e^{i\gamma_j}.$$

At the final index N,

$$S_N^\zeta = \sum_{k=1}^{N} e^{i\gamma_k}, \qquad C_N^\zeta := \frac{1}{N}\sum_{k=1}^{N} S_k^\zeta.$$

The symbol j is only a dummy summation variable inside the running partial sum; the zero positions themselves are indexed by $k = 1, \dots, N$.

Proposition 3.2 (Exact triangular identity)**.** *For every N,*

$$\boxed{C_N^\zeta = \frac{1}{N}\sum_{k=1}^{N}(N-k+1)e^{i\gamma_k} = \sum_{k=1}^{N}\left(1-\frac{k-1}{N}\right)e^{i\gamma_k}.} \tag{3.1}$$

Proof. In $N^{-1}\sum_{k=1}^{N} S_k^\zeta$, the phase $e^{i\gamma_j}$ occurs for $k = j, j+1, \dots, N$, hence exactly $N-j+1$ times. Relabelling the dummy index j as k gives (3.1). □

3.1. **What the triangular weight does and does not do.** Equation (3.1) is the Cesàro $(C,1)$ mean of the partial sums. It is also a Riesz mean of order one relative to the ordinal frequency $\lambda_k = k-1$. For each fixed k,

$$w_{k,N} := 1 - \frac{k-1}{N} \longrightarrow 1.$$

Thus no fixed zero is suppressed in the limiting process. At a fixed N, zeros near the moving endpoint are attenuated; as N grows, that attenuated region moves outward. This observation prevents a common misinterpretation of numerical stabilization: convergence can be fast while contributions continue to arrive from the entire spectrum, because convergence is produced by oscillatory cancellation rather than by permanent truncation of a tail.

The distinction also matters when comparing Riesz and Abel regularizations. The centroid uses the ordinal scale k, whereas an Abel factor $e^{-\varepsilon\gamma_k}$ regularizes in the height variable. Since

$$k \sim \frac{\gamma_k}{2\pi}\log\gamma_k,$$

the two kernels are not identical at the order-one scale. Any passage between them requires a Tauberian or boundary-interchange argument; it is not automatic.

4. Unconditional elimination of the counting self-correlation

Define the linear exact-clock statistic

$$Q_N^{(H)} := \frac{1}{N}\sum_{k=1}^{N}\left(N+\frac{1}{2}-H(\gamma_k)\right)e^{i\gamma_k}. \tag{4.1}$$

By the definition of r_k and (3.1),

$$\boxed{C_N^\zeta = Q_N^{(H)} - \frac{1}{N}\sum_{k=1}^{N} r_k e^{i\gamma_k}.} \tag{4.2}$$

The following theorem is the unconditional Level 1 statement.

Theorem 4.1 (Unconditional Level 1)**.** *For a height $x > 0$ that is not a zero ordinate, define the complete-block sum*

$$A(x) := \sum_{\substack{\gamma \le x \\ \gamma \text{ distinct}}} \nu(\gamma) S_*(\gamma) e^{i\gamma}. \tag{4.3}$$

Then

$$A(x) = o(x \log x) \qquad (x \to \infty). \tag{4.4}$$

Consequently, in the primary ordinal variable,

$$\boxed{\frac{1}{N} \sum_{k=1}^{N} r_k e^{i\gamma_k} = o(1),} \tag{4.5}$$

and therefore

$$\boxed{C_N^{\zeta} - Q_N^{(H)} \longrightarrow 0} \tag{4.6}$$

unconditionally.

Proof. The proof is performed first at complete ordinate blocks and only then transported to the ordinal endpoint N, as prescribed in Remark 2.2. As Stieltjes measures,

$$\mathrm{d}\mathcal{N} = \mathrm{d}H + \mathrm{d}S.$$

Because the jump of $\mathcal{N}$ at a distinct zero ordinate γ is exactly $\nu(\gamma)$, the left side of (4.3) is precisely

$$\begin{aligned} A(x) &= \int_0^x e^{it} S_*(t) \, \mathrm{d}\mathcal{N}(t) \\ &= \int_0^x e^{it} S_*(t) \, \mathrm{d}S(t) + \int_0^x e^{it} H'(t) S(t) \, \mathrm{d}t + O(1). \end{aligned} \tag{4.7}$$

The midpoint convention gives the exact jump identity

$$S_* \, \mathrm{d}S = \frac{1}{2} \, \mathrm{d}(S^2).$$

This remains exact for a jump of any size $\nu(\gamma)$: in particular it covers a multiple zero on the critical line and an off-line functional-equation pair producing two or more copies of the same positive ordinate. Hence

$$A(x) = \frac{1}{2} e^{ix} S(x)^2 - \frac{i}{2} \int_0^x e^{it} S(t)^2 \, \mathrm{d}t + \int_0^x e^{it} H'(t) S(t) \, \mathrm{d}t + O(1). \tag{4.8}$$

Selberg's unconditional second-moment theorem gives

$$\int_0^x S(t)^2 \, \mathrm{d}t \ll x \log\log x, \tag{4.9}$$

so

$$\int_0^x e^{it} S(t)^2 \, \mathrm{d}t = O(x \log\log x).$$

The standard unconditional pointwise estimate $S(x) = O(\log x)$ also gives $e^{ix} S(x)^2 = O(\log^2 x)$.

For the linear term put

$$B(x) := \sum_{\gamma_k \le x} e^{i\gamma_k}, \qquad F(x) := \int_0^x e^{it} S(t) \, \mathrm{d}t, \tag{4.10}$$

where the sum over k counts every upper-half-plane zero with its algebraic multiplicity. Since $\mathrm{d}S = \mathrm{d}\mathcal{N} - H'(t) \, \mathrm{d}t$, Stieltjes integration by parts gives

$$iF(x) = e^{ix} S(x) - B(x) + \int_0^x e^{it} H'(t) \, \mathrm{d}t + O(1). \tag{4.11}$$

Write $e^{i\gamma} = e^{2\pi i\alpha\gamma}$ with $\alpha = (2\pi)^{-1}$. The input used here is exactly Corollary 2 of Ford–Zaharescu [11], not ordinary equidistribution. For every C^2-test function their formula contains the main term $\mathcal{N}(x)\int f$, the secondary term $x\int f g_\alpha$, and an $o(x)$ remainder. With $f(u) = e^{2\pi iu}$, the mean is zero; and $g_\alpha \equiv 0$ because $\alpha = (2\pi)^{-1}$ is non-resonant. Consequently

$$B(x) = o(x) \tag{4.12}$$

unconditionally. The sum $B(x)$ is the complete height sum and counts every zero with algebraic multiplicity. Ordinary Hlawka equidistribution alone would give only $o(\mathcal{N}(x)) = o(x\log x)$ and would not suffice for the later ordinal rate.

Moreover $H'(t) = O(\log t)$ and $H''(t) = O(1/t)$, so

$$\int_0^x e^{it}H'(t)\ \mathrm{d}t = O(\log x),$$

and (4.11) yields

$$F(x) = o(x). \tag{4.13}$$

A further integration by parts gives

$$\int_0^x e^{it}H'(t)S(t)\ \mathrm{d}t = H'(x)F(x) - \int_0^x H''(t)F(t)\ \mathrm{d}t + O(1) \tag{4.14}$$

$$= o(x\log x). \tag{4.15}$$

Indeed, the endpoint is $o(x\log x)$, while $F(t) = o(t)$ and $H''(t) = O(1/t)$ imply $\int_0^x H''(t)F(t)\ \mathrm{d}t = o(x)$. Substitution in (4.8) proves (4.4) for complete blocks.

Now fix an arbitrary ordinal endpoint N. Let the block containing γ_N be $a(\gamma_N)+1,\dots,b(\gamma_N)$. Completing the possibly partial endpoint block changes $\sum_{k\le N} r_k e^{i\gamma_k}$ by at most

$$O(\nu(\gamma_N)(|S_*(\gamma_N)| + \nu(\gamma_N))) = O(\log^2(2+\gamma_N)), \tag{4.16}$$

by (2.11). The completed sum is $A(\gamma_N)$ by (2.8). Therefore

$$\sum_{k=1}^{N} r_k e^{i\gamma_k} = o(\gamma_N\log\gamma_N) + O(\log^2(2+\gamma_N)).$$

Lemma 2.1 gives $N \asymp \gamma_N\log(2+\gamma_N)$, so division by N proves (4.5). Equation (4.6) follows from the exact identity (4.2). □

Remark 4.2. The theorem proves equality of the limiting behavior of C_N^ζ and $Q_N^{(H)}$; it does not, by itself, assert that either sequence converges. Convergence and evaluation of the limiting linear statistic require an additional argument.

5. Exact Riemann–Siegel clock versus Lambert clock

Define the main-clock ordinal defect

$$r_k^{(M)} := k - \frac{1}{2} - M(\gamma_k). \tag{5.1}$$

By (2.15),

$$r_k^{(M)} = r_k + O(\gamma_k^{-1}).$$

The zero-counting estimate $\mathcal{N}(x) \ll x\log x$ gives

$$\sum_{\gamma_k\le x}\frac{1}{\gamma_k} \ll \log^2 x.$$

Therefore, at the ordinal endpoint N,

$$\sum_{k\le N}(r_k^{(M)} - r_k)e^{i\gamma_k} = O(\log^2(2+\gamma_N)) = o(N), \tag{5.2}$$

where the last step follows from Lemma 2.1. Repeated ordinates, whether caused by critical-line multiplicity or by off-line symmetry, are already included in the sum over k.

Corollary 5.1 (Clock equivalence for Level 1)**.** *Replacing $H(\gamma_k)$ by $M(\gamma_k)$ in the linearization of Theorem 4.1 changes the normalized remainder by $o(1)$. Thus the exact Riemann–Siegel clock and the Lambert main clock are asymptotically equivalent for the Level 1 statement.*

For the dataset of $N = 2{,}001{,}052$ ordinates, the exact-clock linear statistic is

$$Q_N^{(H)} = 0.776905882507509 + 0.767853218705326\,i,$$

and the difference between the exact-H and Lambert-M versions is

$$|Q_N^{(H)} - Q_N^{(M)}| = 2.812 \times 10^{-10}.$$

These finite values are numerical evidence only; Corollary 5.1 is the analytic asymptotic statement.

5.1. **The true-zero M-weighted Landau statistic.** The smooth clock can be used as a weight while leaving the ordinates themselves unchanged. This is a different operation from replacing γ_k by the Lambert levels τ_k.

Definition 5.2 (True-zero Landau-weighted statistic)**.** For each ordinal endpoint N, put

$$L_N^{(M)} := \sum_{k=1}^{N} \left(1 - \frac{M(\gamma_k)}{M(\gamma_N)}\right) e^{i\gamma_k}. \tag{5.3}$$

Also define

$$Q_N^{(M)} := \frac{1}{N} \sum_{k=1}^{N} \left(N + \frac{1}{2} - M(\gamma_k)\right) e^{i\gamma_k}. \tag{5.4}$$

Theorem 5.3 (True-zero weighted target)**.** *One has, unconditionally,*

$$L_N^{(M)} - C_N^{\zeta} \longrightarrow 0. \tag{5.5}$$

Consequently, $L_N^{(M)}$ converges if and only if C_N^{ζ} converges, and in that case the two limits are equal.

Proof. Corollary 5.1 gives

$$C_N^{\zeta} - Q_N^{(M)} = o(1). \tag{5.6}$$

Write

$$M_N^{\mathrm{end}} := M(\gamma_N) = N - \frac{1}{2} - r_N^{(M)}. \tag{5.7}$$

The standard pointwise bound for the argument term and (2.15) give

$$M_N^{\mathrm{end}} = N + O(\log(2 + \gamma_N)). \tag{5.8}$$

Let

$$B_N := \sum_{k \le N} e^{i\gamma_k}, \qquad A_N^{(M)} := \sum_{k \le N} M(\gamma_k) e^{i\gamma_k}. \tag{5.9}$$

Direct subtraction of (5.3) and (5.4) gives the exact identity

$$L_N^{(M)} - Q_N^{(M)} = \left(\frac{1}{N} - \frac{1}{M_N^{\mathrm{end}}}\right) A_N^{(M)} - \frac{B_N}{2N}. \tag{5.10}$$

At the fixed non-resonant frequency $x = e$, (4.12) gives

$$B_N = o\left(\frac{N}{\log N}\right), \tag{5.11}$$

because if $\gamma = \gamma_N$, completing the endpoint block changes B_N by at most $\nu(\gamma) = O(\log(2+\gamma))$, while (4.12) gives $o(\gamma)$ for the completed block and Lemma 2.1 gives $N \asymp \gamma_N \log(2+\gamma_N)$. This argument is unchanged if the block is enlarged by critical-line multiplicity or by an off-line functional-equation pair. Abel summation applied to the ordinal sequence $e^{i\gamma_k}$ therefore yields

$$\sum_{k\le N} k e^{i\gamma_k} = o\left(\frac{N^2}{\log N}\right). \tag{5.12}$$

Since

$$M(\gamma_k) = k + O(\log(2+\gamma_k)), \tag{5.13}$$

we obtain, estimating the error term absolutely,

$$A_N^{(M)} = o\left(\frac{N^2}{\log N}\right) + O(N\log N). \tag{5.14}$$

Equations (5.8), (5.11), and (5.14) inserted into (5.10) give

$$L_N^{(M)} - Q_N^{(M)} = o(1). \tag{5.15}$$

Combining this with (5.6) proves (5.5). □

Remark 5.4 (The numerical target). For the run with $N = 2{,}001{,}052$,

$$C_N^{\zeta} = 0.776907119554635 + 0.767844590760710\,i. \tag{5.16}$$

The corresponding exact-clock statistic was

$$Q_N^{(H)} = 0.776905882507509 + 0.767853218705326\,i, \tag{5.17}$$

so

$$\left|C_N^{\zeta} - Q_N^{(H)}\right| = 8.7162 \times 10^{-6}. \tag{5.18}$$

Thus a closed-form candidate obtained from a weighted Landau calculation on the true zeros must be compared with (5.16); the synthetic Lambert centroid belongs to a different problem.

6. The RVM exponential centroid: an unconditional limit

We now study a different object: the exponential walk generated by the synthetic Lambert ordinates τ_k themselves.

Definition 6.1. Let τ_k be given by (2.16). For the running index $k = 1, \dots, N$, define

$$S_k^M := \sum_{j=1}^{k} e^{i\tau_j}, \qquad C_N^M := \frac{1}{N}\sum_{k=1}^{N} S_k^M.$$

Put

$$g_k = \tau_{k+1} - \tau_k, \qquad v_k = e^{i\tau_k}, \qquad d_k = \frac{1}{e^{ig_k} - 1}.$$

The identity

$$v_k = d_k(v_{k+1} - v_k)$$

is exact. Summing it gives the discrete Abel decomposition

$$S_N^M = d_N v_{N+1} - d_1 v_1 - \sum_{k=1}^{N-1} (d_{k+1} - d_k) v_{k+1}. \tag{6.1}$$

Lemma 6.2 (Smooth-clock asymptotics)**.** *As $k \to \infty$,*

$$\tau_k \sim \frac{2\pi k}{\log k}, \qquad g_k \sim \frac{2\pi}{\log k},$$

and

$$d_k = -\frac{i}{g_k} - \frac{1}{2} + O(g_k) = O(\log k), \qquad d_{k+1} - d_k = O(k^{-1}),$$

with a second difference $\Delta^2 d_k = O(k^{-2})$.

Proof. The first two estimates follow from inversion of M and the inverse-function identity

$$\tau'(x) = \frac{1}{M'(\tau(x))}, \qquad M'(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}.$$

The expansion

$$\frac{1}{e^{ix} - 1} = -\frac{i}{x} - \frac{1}{2} + O(x)$$

then yields the estimate for d_k. Differentiating the continuous interpolation $d(x)$ once and twice, using $\tau'(x) \asymp 1/\log x$ and $\tau''(x) = O((x \log x)^{-1})$, gives the difference estimates by the mean-value theorem. □

Lemma 6.3 (Smooth-clock exponential-sum bound)**.** *There is a constant C such that*

$$\left| \sum_{k \le N} e^{i\tau_k} \right| \le C \log N.$$

More generally, interval sums beginning at a sufficiently large index satisfy the corresponding first-derivative bound.

Proof. The continuous inverse $\tau(x)$ has positive monotone derivative

$$\tau'(x) = \frac{2\pi}{\log(\tau(x)/(2\pi))} \asymp \frac{1}{\log x}.$$

For large x, $0 < \tau'(x) < 2\pi$ and its distance from $2\pi\mathbb{Z}$ is bounded below by a constant multiple of $1/\log N$ on $[1, N]$. The Kusmin–Landau first-derivative estimate for exponential sums therefore gives $O(\log N)$. The finite initial segment is absorbed into the constant. □

Theorem 6.4 (Existence and Abel series for the Lambert RVM centroid)**.** *The centroid C_N^M converges. Its limit is*

$$\boxed{C_M = -d_1 e^{i\tau_1} - \sum_{k=1}^{\infty} (d_{k+1} - d_k) e^{i\tau_{k+1}}.} \tag{6.2}$$

The series converges conditionally, and its tail satisfies

$$\sum_{k \ge J} (d_{k+1} - d_k) e^{i\tau_{k+1}} = O\left(\frac{\log J}{J} \right). \tag{6.3}$$

Moreover

$$C_N^M - C_M = O\left(\frac{(\log N)^2}{N} \right).$$

Proof. Let $q_k = d_{k+1} - d_k$. Lemma 6.2 gives $q_k = O(k^{-1})$ and $\Delta q_k = O(k^{-2})$, while Lemma 6.3 gives partial phase sums of order $O(\log k)$. Abel summation therefore yields

$$\sum_{k \ge J} q_k v_{k+1} = O\left(\frac{\log J}{J} \right),$$

which proves convergence and (6.3).

Average (6.1) over the running endpoint $k = 1, \dots, N$. One obtains the exact identity

$$C_N^M = -d_1 v_1 + \frac{1}{N}\sum_{k=1}^{N} d_k v_{k+1} - \sum_{k=1}^{N-1}\left(1 - \frac{k}{N}\right) q_k v_{k+1}. \tag{6.4}$$

A further Abel summation, now with $d_k = O(\log k)$ and the phase-sum bound, gives

$$\frac{1}{N}\sum_{k\le N} d_k v_{k+1} = O\left(\frac{(\log N)^2}{N}\right).$$

The difference between the weighted finite series in (6.4) and the infinite series in (6.2) is the sum of the tail (6.3) and

$$\frac{1}{N}\sum_{k<N} k q_k v_{k+1},$$

which is $O(\log N/N)$ by Abel summation after the smooth asymptotic expansion. Together with the preceding boundary-average estimate this gives the stated $O((\log N)^2/N)$ rate and proves the theorem. □

Remark 6.5. The same strategy applies to the exact smooth clock $H(\widehat{\tau}_k) = k - \frac{1}{2}$: its inverse derivative is again monotone with the same asymptotic scale, so an analogous Abel representation and convergence theorem follow. The closed Lambert formula (2.16), however, makes the M-clock version particularly transparent and directly reproducible.

7. The zeta centroid and the explicit formula

The unconditional theorem 4.1 reduces C_N^{ζ} to the linear statistic $Q_N^{(H)}$. Evaluation by a holomorphic explicit formula is a separate step.

7.1. **What non-resonance at $x = e$ removes.** Landau's formula and its uniform refinements distinguish an integral prime power from a non-resonant fixed value of x [9, 10, 11]. At $x = e$ one has $\Lambda(e) = 0$, so the term of size x supported at an exact prime-power resonance is absent. This fact is essential for the cancellation estimate used in Section 4, but it is a statement at the leading height scale x.

Under RH, the uniform Landau formula of Ford–Zaharescu gives, for fixed $x = e$ and nearest integral prime power $n_e = 3$,

$$\sum_{\gamma_k \le x} e^{i\gamma_k} = -\frac{\log 3}{2\pi\sqrt{e}}\,\frac{e^{ix(1-\log 3)} - 1}{i(1-\log 3)} + O(\log^2 x). \tag{7.1}$$

Equation (7.1) is not an evaluation of the order-one triangular centroid: its error term is larger than the displayed bounded contribution. Its role here is logical. The condition $\Lambda(e) = 0$ removes the exact resonant term of order x; it does not imply that every lower-order arithmetic contribution to a one-sided or Cesàro-regularized value vanishes.

For an arbitrary ordinal endpoint N, complete the ordinate block containing γ_N. Under RH all zeros in that block lie on the critical line, but any algebraic multiplicity is still repeated in the index. Equation (7.1) applied at the complete block, followed by the $O(\nu(\gamma_N)) = O(\log(2+\gamma_N))$ partial-block correction, gives the purely ordinal consequence

$$\sum_{k\le N} e^{i\gamma_k} = O\big(\big(\log(2+N)\big)^2\big) \qquad \text{under RH.} \tag{7.2}$$

No zero is lost in this conversion: a critical-line zero of multiplicity m contributes m identical terms. Off-line blocks are irrelevant to (7.2) because that displayed estimate is invoked under RH; the

unconditional argument of Section 4 instead uses Ford–Zaharescu and the full upper-half-plane multiplicity convention.

7.2. **The symmetric real coordinate.** Let

$$\Psi(u) = \psi(e^u),$$

where ψ is Chebyshev's function. The classical explicit formula may be written, with the usual symmetric regularization,

$$\Psi(u) = e^u - \sum_{\rho}^{\mathrm{reg}} \frac{e^{\rho u}}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - e^{-2u}). \tag{7.3}$$

Differentiating distributionally gives

$$\Psi'(u) = e^u - \sum_{\rho}^{\mathrm{reg}} e^{\rho u} - \frac{e^{-2u}}{1 - e^{-2u}}. \tag{7.4}$$

At $u = 1$, the point $e^u = e$ is not an integer and hence is not a prime power; there is no Chebyshev atom at this point. Therefore

$$\boxed{\sum_{\rho}^{\mathrm{reg}} e^{\rho} = e - \frac{1}{e^2 - 1}.} \tag{7.5}$$

This identity is a classical consequence of the explicit formula. The centroid interpretation needs an additional summability identification.

Hypothesis 7.1 (Symmetric Riesz compatibility at $u = 1$)**.** *The symmetric regularization in* (7.5)*, after pairing ρ and $\bar{\rho}$, agrees with the limit of the ordinal triangular Cesàro real statistic obtained from $Q_N^{(H)}$.*

Theorem 7.2 (Closed real coordinate)**.** *Assume RH and Hypothesis 7.1. Then the zeta centroid converges in real part and*

$$\boxed{\lim_{N\to\infty} \Re C_N^{\zeta} = \frac{1}{2\sqrt{e}}\left(e - \frac{1}{e^2 - 1}\right) = 0.77689426079325358\ldots.} \tag{7.6}$$

Proof. Under RH, every positive zero is $\rho = \frac{1}{2} + i\gamma$, and its conjugate is $\bar{\rho} = \frac{1}{2} - i\gamma$. Thus

$$e^{\rho} + e^{\bar{\rho}} = 2\sqrt{e}\cos\gamma.$$

Pairing the zeros in (7.5), dividing by $2\sqrt{e}$, and applying Hypothesis 7.1 yields the limit of the real triangular statistic. The unconditional Level 1 theorem transfers the same limit to C_N^{ζ}. □

Remark 7.3 (What is classical and what is specific). Equation (7.5) is not presented as a new explicit formula. The specific assertion is the identification of its value with the real coordinate selected by the ordinal triangular centroid. That identification is exactly the content isolated in Hypothesis 7.1; it should not be conflated with the classical bilateral regularization itself.

8. One-sided transform and the full complex value

The imaginary coordinate cannot be recovered by pairing $+\gamma$ and $-\gamma$, because

$$e^{i\gamma} + e^{-i\gamma} = 2\cos\gamma.$$

It requires a one-sided projection onto positive ordinates.

Fix $x_0 \in (0, \gamma_1)$. For $\sigma > 1$, define

$$s_\sigma(t) = \frac{1}{\pi}\Im \operatorname{Log}\zeta(\sigma + it).$$

By the Euler product,

$$s_\sigma(t) = -\frac{1}{\pi}\sum_{n\geq 2}\frac{\Lambda(n)}{n^\sigma \log n}\sin(t\log n). \tag{8.1}$$

The one-sided triangular transform gives

$$\mathcal{P}_{\infty,\sigma}(x_0) = \frac{ie^{ix_0}}{2\pi}\sum_{n\geq 2}\frac{\Lambda(n)}{n^\sigma \log n}\left[\frac{n^{ix_0}}{1+\log n} - \frac{n^{-ix_0}}{1-\log n}\right]. \tag{8.2}$$

For $\sigma > 1$ this is an absolutely justified rearrangement. Separating the slowly decaying $n = 2$ term in the second branch yields the equivalent horizontal-$\operatorname{Log}\zeta$ formula

$$\begin{aligned}\mathcal{P}_{\infty,\sigma}(x_0) = \frac{ie^{ix_0}}{2\pi}\Bigg\{ &\int_0^\infty e^{-u}\operatorname{Log}\zeta(\sigma+u-ix_0)\,\mathrm{d}u \\ &+ \int_0^\infty e^{u}\left[\operatorname{Log}\zeta(\sigma+u+ix_0) - 2^{-(\sigma+u+ix_0)}\right]\mathrm{d}u \\ &- \frac{2^{-\sigma-ix_0}}{1-\log 2}\Bigg\}.\end{aligned} \tag{8.3}$$

The two denominators $1 \pm \log n$ are the analytic record of the one-sided frequency split. Although e itself is not a prime power, the one-sided projection contains nonlocal prime contributions; therefore the full complex centroid is not purely archimedean merely because the symmetric atom at $x = e$ vanishes.

Remark 8.1 (The proposed arithmetic-free route). There is a valid conditional implication but not a theorem of cancellation. If one could prove independently that

$$\lim_{\sigma\downarrow 1/2}\mathcal{P}_{\infty,\sigma}(x_0) = 0, \tag{8.4}$$

then the full constant in Theorem 8.3 would reduce to the smooth term $\mathcal{B}(x_0)$, which is built from the gamma factor and its derivatives. However, $\Lambda(e) = 0$ does not imply (8.4). In the present one-sided representation the prime contribution is the entire series (8.2), not a single atom at $n = e$. Moreover, the reproducibility evaluation at $x_0 = 1$ gives a nonzero boundary contribution

$$\mathcal{P}(1) = 0.395112242469534 + 0.621692567335957\,i. \tag{8.5}$$

Thus a reduction of the full complex centroid to a small list of archimedean constants would require a new cancellation identity not supplied by fixed-frequency non-resonance. The symmetric real coordinate is different: there the bilateral explicit formula at $u = 1$ has no Chebyshev atom and yields the closed value in Theorem 7.2.

The smooth contribution is

$$\mathcal{B}(x_0) = e^{ix_0}\big(H(x_0) + iH'(x_0)\big) + i\int_{x_0}^\infty e^{it}H''(t)\,\mathrm{d}t. \tag{8.6}$$

Hypothesis 8.2 (One-sided boundary interchange)**.** *Assume that the finite ordinal triangular transform of the argument term may be passed to the critical-line boundary in the order*

$$N \to \infty, \qquad \sigma \downarrow \frac{1}{2},$$

so that the limiting one-sided arithmetic term equals

$$\lim_{\sigma\downarrow 1/2}\mathcal{P}_{\infty,\sigma}(x_0),$$

with the branch of $\operatorname{Log}\zeta$ *obtained by continuation from* $\Re s > 1$*. Equivalently, the relevant finite* $S(t)$ *transform and the regularized horizontal-*$\operatorname{Log}\zeta$ *transform have the same boundary value.*

This hypothesis is the precise unresolved summability/contour interchange. It is deliberately separated from RH: RH controls the horizontal position of zeros, while Hypothesis 8.2 controls the equality of two summation procedures.

Theorem 8.3 (Conditional full complex centroid)**.** *Assume RH and Hypothesis 8.2. Then C_N^ζ converges and, for every admissible $x_0 \in (0, \gamma_1)$,*

$$\boxed{C_\infty^\zeta = \mathcal{B}(x_0) + \lim_{\sigma \downarrow 1/2} \mathcal{P}_{\infty,\sigma}(x_0).} \tag{8.7}$$

For $x_0 = 1$, numerical evaluation of the zero-free right-hand side gives

$$\boxed{C_\infty^\zeta = 0.77689426079325\ldots + 0.76785566886455\ldots\, i.} \tag{8.8}$$

Its real part agrees with the closed form (7.6)*.*

Proof. The exact Stieltjes decomposition behind Theorem 4.1 reduces the centroid to a smooth term and a linear transform of $S(t)$, while the quadratic term contributes $o(1)$ by Selberg's second moment. The smooth term tends to (8.6). For $\sigma > 1$, inserting (8.1) into the linear transform and integrating each sine mode gives (8.2), equivalently (8.3). Hypothesis 8.2 authorizes passage to the critical-line boundary, and RH identifies the holomorphic complete-zero phase with the pure ordinate phase. The unconditional Level 1 theorem then transfers the result to the original centroid. The numerical value follows from direct quadrature of (8.6) and (8.3); the accompanying Python script reproduces it without a table of zeta zeros. □

For reference, the reproducibility calculation at $x_0 = 1$ gives approximately

$$\mathcal{B}(1) = 0.38178201832363 + 0.14616310152859\, i,$$

$$\mathcal{P}(1) = 0.395112242469534 + 0.621692567335957\, i,$$

whose sum is (8.8).

8.1. **The holomorphic obstruction.** The role of RH can be stated without reference to any particular explicit formula.

Proposition 8.4 (No universal holomorphic ordinate-phase kernel)**.** *There is no holomorphic function K on an open set containing a two-dimensional portion of the critical strip such that*

$$\Im K(\sigma + it) = \sin t$$

independently of σ throughout that open set.

Proof. The imaginary part of a holomorphic function is harmonic. But

$$\Delta \sin t = \frac{\partial^2}{\partial \sigma^2} \sin t + \frac{\partial^2}{\partial t^2} \sin t = -\sin t,$$

which is not identically zero on any open set. □

Under RH, however,

$$K(s) = e^{s-1/2}$$

satisfies $K(\frac{1}{2} + i\gamma) = e^{i\gamma}$. Thus RH is precisely what allows the standard holomorphic complete-zero weight to specialize to the pure ordinate phase at every nontrivial zero.

9. The off-critical-line defect

Suppose $\rho = \frac{1}{2} + \delta + i\gamma$ is paired by the functional equation with $\frac{1}{2} - \delta + i\gamma$. The pure ordinate statistic assigns the pair the weight

$$2e^{i\gamma},$$

whereas the holomorphic complete-zero statistic gives

$$e^{\delta+i\gamma} + e^{-\delta+i\gamma} = 2\cosh\delta\, e^{i\gamma}.$$

The completed-minus-ordinate pair excess at the real boundary frequency $u = 1$ is therefore

$$2(\cosh\delta - 1)e^{i\gamma}. \tag{9.1}$$

For a fixed functional-equation pair, the two ordinal triangular weights both tend to one as $N \to \infty$, regardless of how equal ordinates are ordered at finite N. Hence its completed-minus-ordinate contribution tends to (9.1). We do not identify the finite-N paired difference with a single common triangular weight, because the two members of a tied pair may occupy different ordinal positions.

Two features are fundamental. First, Riesz tapering does not erase a hypothetical off-line zero at a fixed large height. The reason a zero extremely close to the critical line is hard to detect is instead the quadratic sensitivity

$$2(\cosh\delta - 1) = \delta^2 + O(\delta^4). \tag{9.2}$$

For example, $\delta = 10^{-6}$ produces a limiting pair amplitude of approximately 10^{-12}.

Second, the limiting pair coefficients $2(\cosh\delta - 1)$ are nonnegative but the phases $e^{i\gamma}$ are not. Consequently, vanishing of a regularized real-boundary excess at the isolated frequency $u = 1$ would not force every coefficient to vanish: a positive discrete measure may have zero Fourier transform at one real frequency.

The cancellation issue at the real boundary frequency $u = 1$ is resolved in a different way in Section 10: moving the *same centroid family* to the positive imaginary axis converts the horizontal defect into a positive Laplace sum.

10. A centroid-family equivalence to the Riemann hypothesis

This section starts from the centroid itself and changes only its frequency parameter. The same triangular weights are used on both sides of the comparison. No smoothing of the zero sequence, contour evaluation, or boundary-interchange hypothesis enters the equivalence.

10.1. **Holomorphic realization of the ordinal centroid.**

For $u \in \mathbb{C}$ define

$$C_N^{\zeta}(u) := \frac{1}{N}\sum_{k=1}^{N} S_k^{\zeta}(u) = \sum_{k=1}^{N} w_{k,N} e^{iu\gamma_k}, \qquad S_k^{\zeta}(u) := \sum_{j=1}^{k} e^{iu\gamma_j}, \qquad w_{k,N} = 1 - \frac{k-1}{N}. \tag{10.1}$$

Thus $C_N^{\zeta}(1) = C_N^{\zeta}$. The real point $u = 1$ belongs to the boundary of the half-plane used below.

Theorem 10.1 (Holomorphic centroid in the upper frequency half-plane)**.** *For every u with $\Im u > 0$, the limit*

$$C^{\zeta}(u) := \lim_{N\to\infty} C_N^{\zeta}(u) \tag{10.2}$$

exists and equals the absolutely convergent series

$$\boxed{C^{\zeta}(u) = \sum_{k=1}^{\infty} e^{iu\gamma_k}.} \tag{10.3}$$

For every compact $K \subset \{u : \Im u > 0\}$,

$$\sup_{u\in K} |C_N^{\zeta}(u) - C^{\zeta}(u)| = O_K(N^{-1}). \tag{10.4}$$

Consequently C^{ζ} is holomorphic on $\Im u > 0$.

Proof. Let $y_K = \inf_{u\in K} \Im u > 0$. Then

$$|e^{iu\gamma_k}| \le e^{-y_K\gamma_k}. \tag{10.5}$$

The Riemann–von Mangoldt estimate $\mathcal{N}(x) \ll x \log x$ implies

$$\sum_{k\ge1} e^{-y_K\gamma_k} < \infty, \qquad \sum_{k\ge1} k e^{-y_K\gamma_k} < \infty. \tag{10.6}$$

Using (10.1) and (10.3),

$$C_N^{\zeta}(u) - C^{\zeta}(u) = -\frac{1}{N}\sum_{k\le N}(k-1)e^{iu\gamma_k} - \sum_{k>N} e^{iu\gamma_k}. \tag{10.7}$$

The first term is bounded by

$$\frac{1}{N}\sum_{k\le N}(k-1)e^{-y_K\gamma_k} \le \frac{1}{N}\sum_{k\ge1} k e^{-y_K\gamma_k},$$

and, since $k/N \ge 1$ for $k > N$, the second term satisfies

$$\sum_{k>N} e^{-y_K\gamma_k} \le \frac{1}{N}\sum_{k>N} k e^{-y_K\gamma_k}.$$

Thus

$$\sup_{u\in K} |C_N^{\zeta}(u) - C^{\zeta}(u)| \le \frac{2}{N}\sum_{k\ge1} k e^{-y_K\gamma_k} = O_K(N^{-1}),$$

which proves (10.4). Local uniform convergence gives holomorphy by Weierstrass' theorem. Repeated ordinates cause no change: they appear as distinct ordinal slots k, and the convergent majorant counts every such slot. □

The theorem is an interior statement. It does not identify the observed Riesz behavior at $u = 1$ with an Abel or nontangential boundary value of $C^{\zeta}(u)$.

10.2. **The completed-zero centroid.** Write the positive zeros as $\rho_k = \beta_k + i\gamma_k$, with the same multiplicities and ordinate ordering as in (10.1). Let

$$\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\!\left(\frac{s}{2}\right)\zeta(s), \tag{10.8}$$

whose zeros are precisely the nontrivial zeros of ζ, counted with multiplicity [2, 3, 6]. Define

$$C_N^{\xi}(u) := \sum_{k=1}^{N} w_{k,N} e^{u(\rho_k - 1/2)}. \tag{10.9}$$

For $\Im u > 0$ its limit is

$$\boxed{C^{\xi}(u) := \lim_{N\to\infty} C_N^{\xi}(u) = \sum_{\substack{\rho \\ \Im\rho>0}} e^{u(\rho-1/2)}.} \tag{10.10}$$

The finite value in (10.9) can depend on the arbitrary ordering of zeros sharing exactly the same ordinate, but its limit does not. This is the only place where the horizontal locations inside one fixed ordinate block can make the finite ordering visible: the ordinate centroid $C_N^{\zeta}(u)$ itself is insensitive to that ordering because all members of the block have the same γ. If RH holds, a

multiplicity-m zero on the line contributes m identical completed terms. If RH fails, an off-line pair $\beta + i\gamma$, $1 - \beta + i\gamma$, each with multiplicity m, occupies $2m$ ordinal positions at the same γ; after the complete pair is included its contribution is paired exactly as below. The convergence is absolute and locally uniform. Indeed, on a compact K in the upper half-plane,

$$|e^{u(\rho-1/2)}| \le e^{\frac{1}{2}\sup_{u\in K}|\Re u|} e^{-y_K\gamma}, \tag{10.11}$$

and the same argument as in Theorem 10.1 gives an $O_K(N^{-1})$ triangular-to-series error. Hence C^ξ is holomorphic for $\Im u > 0$.

Remark 10.2 (Interior notation and the real boundary). The notation $C^\xi(u)$ is used only for $\Im u > 0$, where the series in (10.10) converges absolutely. In particular, no quantity $C^\xi(1)$ is defined by that series in this paper. The conditional real-boundary limit C^ζ_∞ obtained in Theorem 8.3 is a different object, defined through the ordinal centroid together with the stated boundary-interchange hypothesis. The RH equivalence below is entirely interior and does not use that hypothesis.

10.3. **Positive imaginary-axis defect and RH.** Functional-equation symmetry pairs

$$\frac{1}{2} + \delta_\rho + i\gamma \quad\longleftrightarrow\quad \frac{1}{2} - \delta_\rho + i\gamma, \qquad \delta_\rho = \beta - \frac{1}{2}. \tag{10.12}$$

At $u = iy$, $y > 0$, one simple off-line pair contributes $2e^{-y\gamma}$ to $C^\zeta(iy)$ and $2\cos(y\delta_\rho)e^{-y\gamma}$ to $C^\xi(iy)$. A pair of algebraic multiplicity m contributes m times these quantities. Critical-line multiplicities contribute equally to the two series and hence give zero defect. Therefore

$$\boxed{\mathcal{D}(y) := C^\zeta(iy) - C^\xi(iy) = 2\sum_{\substack{\beta>1/2\\ \gamma>0}} \left(1 - \cos\left(y\left(\beta - \frac{1}{2}\right)\right)\right) e^{-y\gamma} \ge 0.} \tag{10.13}$$

The series converges absolutely for every $y > 0$; the sum over $\beta > 1/2$ counts algebraic multiplicity. Thus every off-line upper-half-plane functional-equation pair is represented exactly once on the right, with its multiplicity, while the ordinate series on the left contains both positive-ordinate members and therefore the same total ordinal count.

Theorem 10.3 (Centroid-family equivalence with RH)**.** *For every fixed y_0 with $0 < y_0 \le 4\pi$,*

$$\boxed{\text{RH} \iff \mathcal{D}(y_0) = 0 \iff C^\zeta(iy_0) = C^\xi(iy_0).} \tag{10.14}$$

In particular,

$$\boxed{\text{RH} \iff C^\zeta(i) = C^\xi(i).} \tag{10.15}$$

Proof. If RH holds, $\delta_\rho = 0$ for every nontrivial zero, so every term in (10.13) vanishes.

Conversely, suppose $\mathcal{D}(y_0) = 0$. Equation (10.13) is a convergent sum of nonnegative terms, so each term is zero. Hence

$$y_0\left(\beta - \frac{1}{2}\right) \in 2\pi\mathbb{Z} \tag{10.16}$$

for every zero with $\beta > 1/2$. Every nontrivial zero lies in the open strip $0 < \beta < 1$ [2, 6]; therefore $|\beta - 1/2| < 1/2$. Since $0 < y_0 \le 4\pi$,

$$\left|y_0\left(\beta - \frac{1}{2}\right)\right| < 2\pi. \tag{10.17}$$

The only multiple of 2π in this open interval is 0, so $\beta = 1/2$ for every nontrivial zero. This is RH. □

Corollary 10.4 (RVM-residual form of the RH criterion)**.** *For $y > 0$ define*

$$\mathcal{R}_{\mathrm{RVM}}(y) := \sum_{k\geq 1} e^{-y\tau_k}\left(e^{-y(\gamma_k-\tau_k)} - 1\right) = C^{\zeta}(iy) - \sum_{k\geq 1} e^{-y\tau_k}, \tag{10.18}$$

$$\mathcal{A}_{\mathrm{RVM}}(y) := C^{\xi}(iy) - \sum_{k\geq 1} e^{-y\tau_k}. \tag{10.19}$$

Both series are absolutely convergent. The index k is the actual upper-half-plane zero index from Section 2: if a positive ordinate is repeated because of algebraic multiplicity or an off-line functional-equation pair, the same γ is repeated in the left series and the subsequent Lambert levels are shifted accordingly. There is exactly one τ_k for each actual ordinal slot. If RH fails, these repeated positive ordinates insert additional ordinal slots relative to an RH-only enumeration and therefore shift every subsequent pairing with τ_k. Accordingly $\gamma_k - \tau_k$ is always an exact ordinally paired *Lambert residual; off RH it should not be interpreted as the displacement obtained by keeping the hypothetical RH indexing fixed. For every $y > 0$,*

$$\boxed{\mathcal{R}_{\mathrm{RVM}}(y) \geq \mathcal{A}_{\mathrm{RVM}}(y).} \tag{10.20}$$

Moreover, for every fixed y_0 with $0 < y_0 \leq 4\pi$,

$$\boxed{\mathrm{RH} \iff \mathcal{R}_{\mathrm{RVM}}(y_0) = \mathcal{A}_{\mathrm{RVM}}(y_0).} \tag{10.21}$$

Equivalently,

$$\boxed{\mathrm{RH} \iff \sum_{k\geq 1} e^{-y_0\tau_k}\left(e^{-y_0(\gamma_k-\tau_k)} - 1\right) = C^{\xi}(iy_0) - \sum_{k\geq 1} e^{-y_0\tau_k}.} \tag{10.22}$$

Proof. Immediate from Theorem 10.3, since by construction

$$\mathcal{R}_{\mathrm{RVM}}(y) - \mathcal{A}_{\mathrm{RVM}}(y) = C^{\zeta}(iy) - C^{\xi}(iy) = \mathcal{D}(y). \tag{10.23}$$

Thus $\mathcal{R}_{\mathrm{RVM}}(y) \geq \mathcal{A}_{\mathrm{RVM}}(y)$, and equality at any fixed $0 < y_0 \leq 4\pi$ is equivalent to $\mathcal{D}(y_0) = 0$, which is equivalent to RH by Theorem 10.3. □

The Lambert clock cancels algebraically from the difference in (10.23). Its role is therefore not to supply an independent proof of RH, but to provide an explicit smooth reference that rewrites the positive centroid defect as a statement about the observed vertical residuals $\gamma_k - \tau_k$. At the real frequency $u = 1$ the paired residual is a complex cancellation problem; on the positive imaginary axis the damped version is ordered by the nonnegative horizontal defect, and equality is exactly the RH criterion above.

The logical point is precise: the criterion starts from the same triangular centroid and embeds it in the complex-frequency family (10.1). It does not assert that the observed boundary value at $u = 1$ alone is an RH criterion. The latter boundary identification remains the separate Riesz/Abel problem isolated in Sections 3 and 8.

11. Fujii's horizontal moment and the spectral zeta function

Define the horizontal second-moment counting function

$$A_2(x) = \sum_{\substack{\beta>1/2\\ 0<\gamma\leq x}} \left(\beta - \frac{1}{2}\right)^2. \tag{11.1}$$

Here zeros with $\beta > 1/2$ are counted with algebraic multiplicity. The cutoff is deliberately a *complete height cutoff*: for an off-line pair at ordinate γ, the member with $\beta > 1/2$ enters together with all multiplicity copies at that ordinate. We do not identify $A_2(x)$ with a partially truncated ordinal sum inside an ordinate block, because such a truncation would depend on the arbitrary ordering of

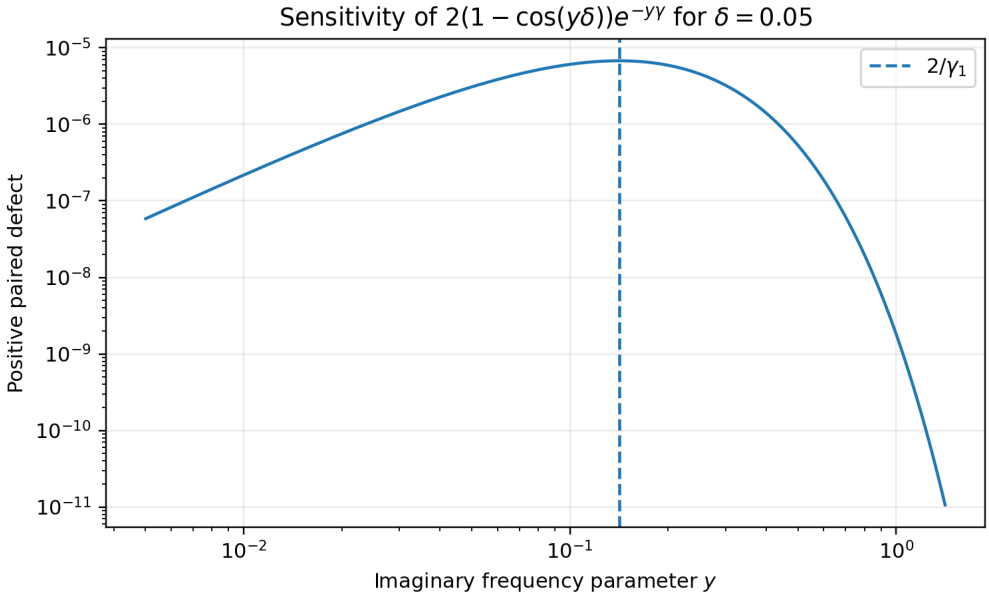


FIGURE 5. Positive paired contribution to $\mathcal{D}(y)$ for the illustrative values $\delta_\rho = 0.05$ and $\gamma = \gamma_1$: $2(1-\cos(y\delta_\rho))e^{-y\gamma}$. In the small-$y\delta_\rho$ regime the leading approximation is $y^2\delta_\rho^2 e^{-y\gamma}$, whose maximum occurs near $y = 2/\gamma$. The figure illustrates numerical sensitivity only; Theorem 10.3 is exact.

horizontally distinct zeros sharing the same γ. Whenever the endpoint is tied to the centroid index, the block containing γ_N is completed first, exactly as in Remark 2.2.

Fujii proved an identity of the form

$$\widetilde{S}_2(x) = A_2(x) + O(\log x), \tag{11.2}$$

where $\widetilde{S}_2$ is an explicitly normalized second antiderivative quantity built from $S(t)$ [20]. Bondarenko–Ivić–Saksman–Seip use this relation in their study of

$$G(s) = \sum_{\gamma>0} \gamma^{-s},$$

showing unconditional meromorphic continuation to $\Re s > -1$ and relating further continuation to the horizontal distribution of zeros [22].

Our defect has the expansion

$$2(\cosh\delta - 1) = \delta^2 + \frac{\delta^4}{12} + \frac{\delta^6}{360} + \cdots.$$

Since $0 \le \delta < 1/2$ for nontrivial zeros,

$$1 \le \frac{2(\cosh\delta - 1)}{\delta^2} \le 8\left(\cosh\frac{1}{2} - 1\right) = 1.02100772\ldots.$$

Thus Fujii's $A_2(x)$ is the leading positive mass underlying the exact horizontal defect. What the centroid needs, however, is not merely $A_2(x)$ but its oscillatory transform at frequency one:

$$\int e^{it}\, \mathrm{d}A_2(t).$$

A zero-density or positive-mass estimate does not imply that this transform vanishes.

For $\Re s > 1$ define

$$H_2(s) = \sum_{\beta>1/2,\gamma>0} \frac{(\beta-1/2)^2}{\gamma^s}.$$

For $\Re z > 0$, the Laplace transform of the horizontal mass has the Mellin representation

$$F_2(z) := \sum_{\beta>1/2,\gamma>0} (\beta-\tfrac{1}{2})^2 e^{-z\gamma} = \frac{1}{2\pi i}\int_{(c)} \Gamma(s) z^{-s} H_2(s)\, \mathrm{d}s, \qquad c>1. \tag{11.3}$$

The desired pure Fourier value is a boundary limit $z = \varepsilon - i$, $\varepsilon \downarrow 0$. At that boundary the exponential decay of $\Gamma(c+it)$ is cancelled on one half of the contour by the factor $(-i)^{-s}$, which explains why the boundary is analytically more delicate than the damped Laplace regime. The Riesz kernel of the centroid does not remove this issue: it is a moving ordinal taper, not an exponential height damping.

This establishes the conceptual end point of the Fujii route at the real boundary frequency $u = 1$. It identifies the exact positive horizontal mass responsible for the failure of the holomorphic ordinate projection, but a positive-mass estimate alone does not determine its oscillatory Fourier value there.

12. Numerical values and reproducibility

The analytic arguments above are independent of the numerical observations. This section records the finite values that motivated the analysis and the independent checks supplied with the manuscript.

12.1. **Finite true-zero and Lambert centroids.** For the dataset with

$$N = 2{,}001{,}052, \qquad \gamma_N \approx 1{,}132{,}490.6587144111,$$

the finite centroids are

$$C_N^{\zeta} = 0.776907119554635 + 0.767844590760710\, i,$$
$$C_N^{\mathrm{Lamb}} = 0.041570219277352 + 1.757541749592316\, i.$$

The direct trajectory average and the triangular weighted formula agree to approximately 1.7×10^{-13} for zeta and 4.9×10^{-14} for the Lambert clock.

The conditional limiting value supplied by Theorem 8.3 is

$$C_{\infty}^{\zeta} = 0.77689426079325\ldots + 0.76785566886455\ldots i.$$

At the available N, the real-coordinate discrepancy from the closed form is approximately

$$1.29\times10^{-5}.$$

No finite-N observation is used as a premise in Theorem 4.1, Theorem 6.4, or the conditional explicit-formula derivations.

12.2. **Lambert-clock reproduction from the closed inverse.** The accompanying script `verify_rvm_centroid.py` reconstructs (2.16) directly rather than reading the `zLamb.bin`. At $N = 2{,}001{,}052$ it obtains

$$C_N^{M} = 0.041570227040394 + 1.757541761467938\, i,$$

with the Abel decomposition

$$-d_1 e^{i\tau_1} = 6.031991313792247 + 2.977851908778479\, i,$$
$$\frac{1}{N}\sum_{k\le N} d_k e^{i\tau_{k+1}} = -1.7006454064\times10^{-6} + 1.3853445067\times10^{-7} i,$$
$$-\sum_{k<N}\left(1-\frac{k}{N}\right)\Delta d_k e^{i\tau_{k+1}} = -5.990419386106493 - 1.220310285845148\, i.$$

The direct and Abel reconstructions differ by approximately 1.6×10^{-13}. The small difference from the Lambert centroid reflects the finite numerical generation of the smooth sequence rather than a change in definition.

12.3. **Zero-free evaluation of the conditional zeta constant.** The script `verify_zeta_conditional_constant.py` evaluates (8.6) and (8.3) at $x_0 = 1$ without reading a zeta-zero table. It gives

$$\begin{aligned} \mathcal{B}(1) &= 0.38178201832363 + 0.14616310152859\, i, \\ \mathcal{P}(1) &= 0.395112242469534 + 0.621692567335957\, i, \\ \mathcal{B}(1) + \mathcal{P}(1) &= 0.77689426079317 + 0.76785566886455\, i. \end{aligned}$$

The independently evaluated real part differs from the closed expression (7.6) by about 9×10^{-14} at the chosen numerical precision and tail truncation.

12.4. **Figures from the numerical tables.** The figures below are generated from the full-data tables and the reproducibility outputs listed in Appendix A. Their source CSV files and the plotting script are included in the accompanying package.

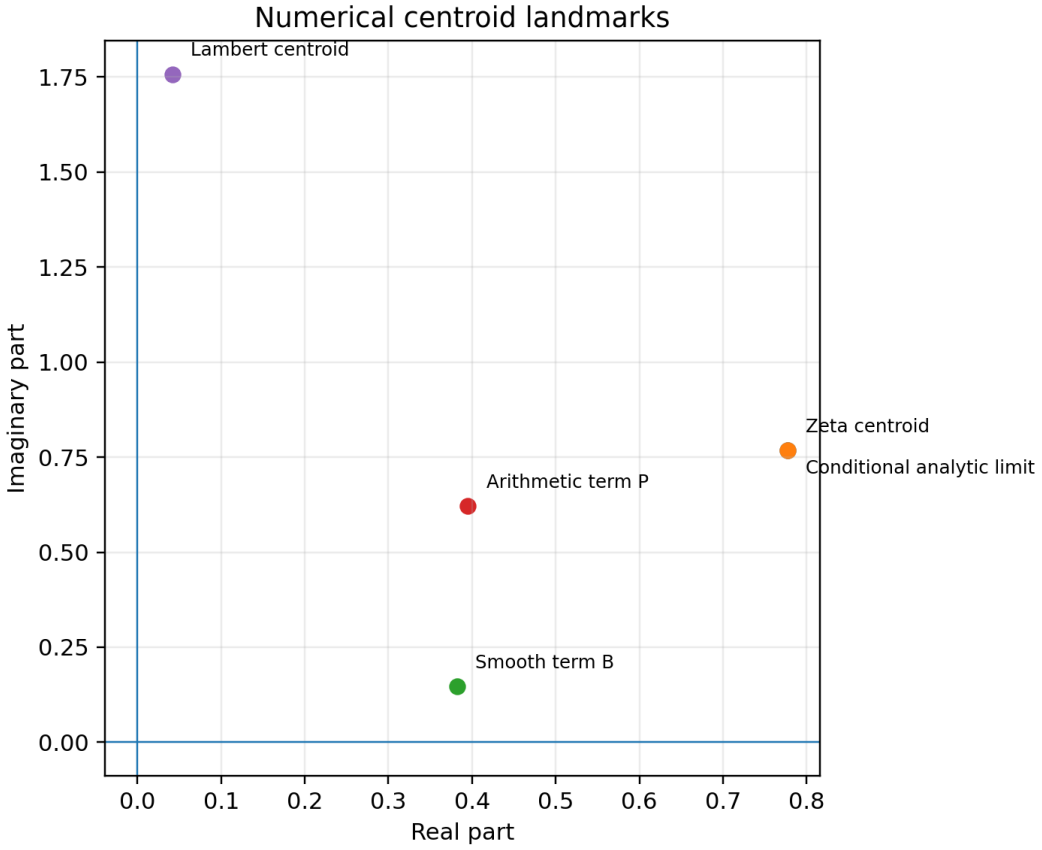


FIGURE 6. Finite and analytic complex landmarks: the zeta centroid, the Lambert centroid, the conditional analytic limit C_∞^ζ, and its smooth/arithmetic decomposition $\mathcal{B} + \mathcal{P}$.

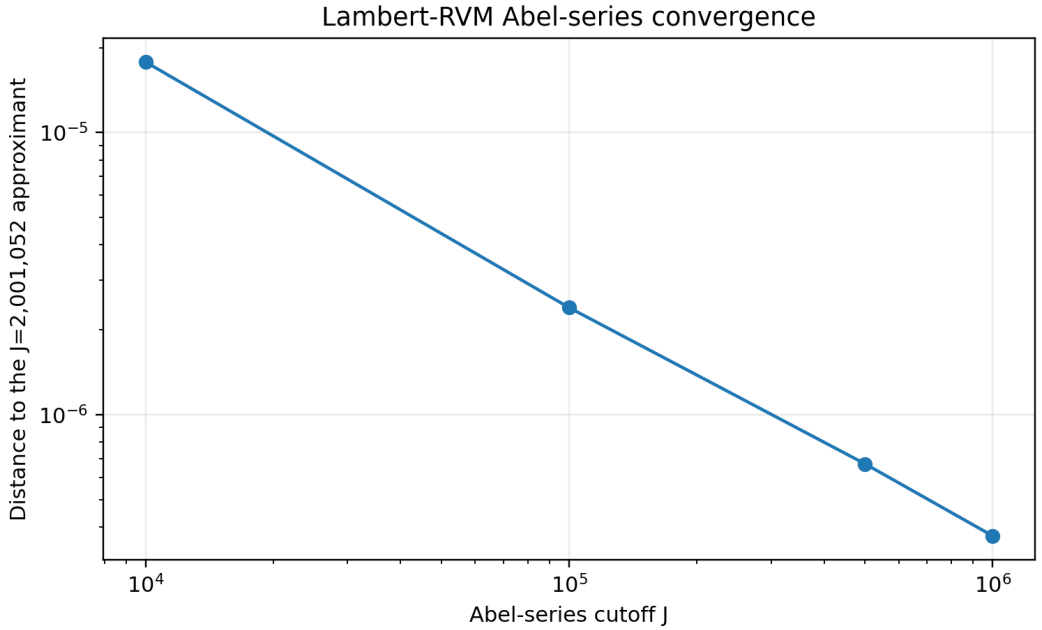


FIGURE 7. Convergence of the Lambert-RVM unweighted Abel-series approximants recorded by `verify_rvm_centroid.py`. The ordinate is the distance from the largest cutoff $J = 2{,}001{,}052$.

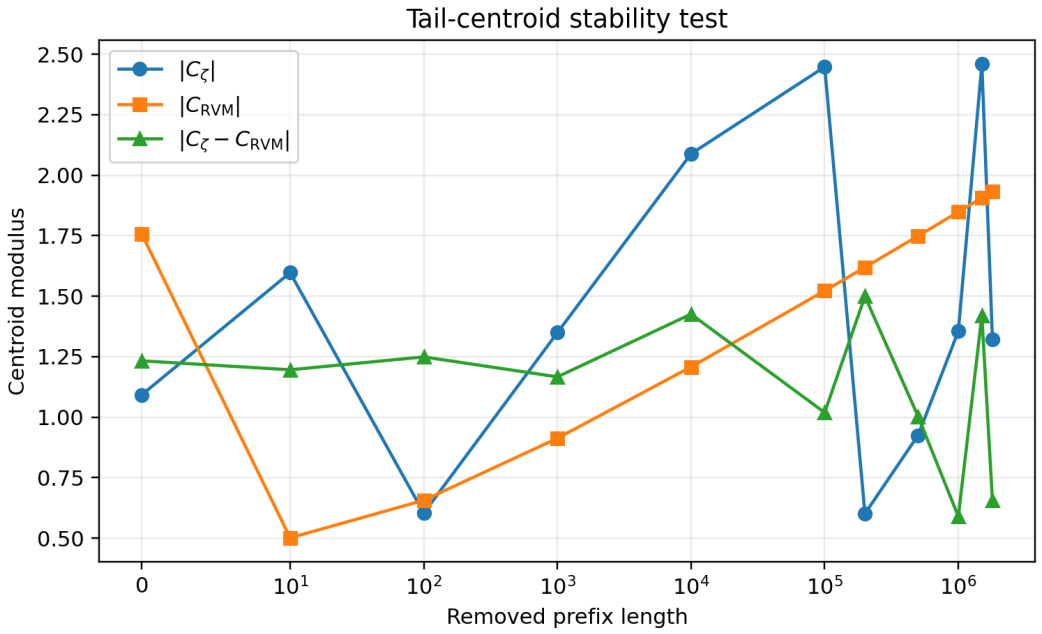


FIGURE 8. Tail-centroid test after deleting increasing prefixes of the true-zero and synthetic RVM sequences. The triangular taper does not permanently protect a finite prefix; the tail statistics remain of order one.

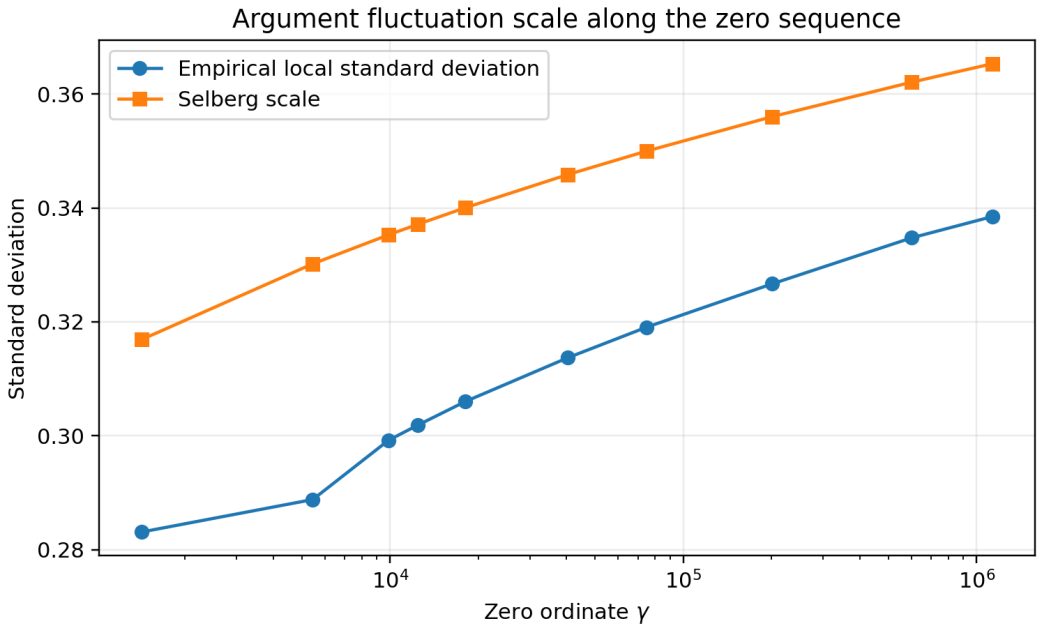


FIGURE 9. Local standard deviation of the midpoint counting defect along the zero sequence compared with the Selberg scale used in the Level 1 analysis.

13. STATUS OF THE MAIN STATEMENTS

In the last column, "None means that no conjectural or additional hypothesis is assumed. Published unconditional results used as inputs—for example Corollary 2 of Ford–Zaharescu in the Level 1 estimate—are cited in the relevant proofs and are not hypotheses of the statements.

Statement	Status	Required hypothesis
Exact triangular centroid identity	Exact finite theorem	None
Unconditional Level 1, $C_N^{\zeta} - Q_N^{(H)} \to 0$	Proved	None
H versus M equivalence in Level 1	Proved	None
True-zero weighted statistic $L_N^{(M)} - C_N^{\zeta} \to 0$	Proved	None
Lambert RVM centroid convergence and Abel series	Proved	None
Regularized bilateral identity at $u = 1$	Classical explicit formula	None for the regularized complete-zero identity
Closed real coordinate of the ordinal zeta centroid	Conditional theorem	RH + symmetric Riesz compatibility
Full complex limit C_{∞}^{ζ}	Conditional theorem	RH + one-sided boundary interchange
Arithmetic-free full complex value from $\Lambda(e) = 0$ alone	Not justified	Non-resonance removes only the leading prime-power atom
Off-line horizontal defect formula	Exact comparison	Functional-equation pairing
Holomorphic centroid $C^{\zeta}(u)$ for $\Im u > 0$	Proved	None
Centroid-family equality $C^{\zeta}(iy) = C^{\xi}(iy)$, $0 < y \le 4\pi$	Equivalent to RH	None
Boundary identification at $u = 1$	Separate summability problem	Riesz/Abel compatibility

14. Conclusions

The triangular exponential centroid admits a sharper analytic decomposition than is visible from its geometric definition. The rank-dependent coefficient can be separated into the exact Riemann–Siegel clock and the ordinal defect r_k; over complete multiplicity blocks the latter aggregates exactly to the midpoint argument term $S_*(\gamma)$. This self-correlation is negligible after normalization unconditionally. This converts the original centroid into a linear statistic without assuming RH.

The smooth RVM control is analytically complete: the Lambert inverse gives an exact model sequence, its exponential endpoint obeys a first-derivative cancellation estimate, and a discrete Abel transform yields a convergent series for the centroid limit. The triangular kernel is thereby seen correctly as a moving ordinal Riesz taper. It attenuates the finite upper edge but does not permanently suppress any fixed part of the spectrum.

The same smooth function M can instead be used only as a weight on the true ordinates. Theorem 5.3 proves that this true-zero Landau-weighted statistic differs from C_N^{ζ} by $o(1)$. Its limiting target is therefore the zeta centroid, not the synthetic Lambert centroid. This removes a possible ambiguity between two constructions that share the same clock but not the same ordinates.

For the zeta centroid, the classical explicit formula at the non-prime-power frequency $x = e$ fixes a symmetric regularized complete-zero value. Under RH and the stated compatibility of that regularization with the ordinal triangular summation, this yields the closed real coordinate

$$\frac{1}{2\sqrt{e}}\left(e - \frac{1}{e^2 - 1}\right).$$

The full complex value requires a one-sided boundary transform. Its arithmetic part is naturally expressed through horizontal integrals of $\operatorname{Log}\zeta$ and contains the two branches $1 \pm \log n$. The absence of an exact prime-power resonance at $x = e$ removes the leading Landau atom but does not annihilate this order-one one-sided arithmetic transform. The remaining mathematical issue is

therefore not hidden: it is the boundary-interchange theorem that identifies this analytic one-sided regularization with the ordinal Cesàro limit.

Off the critical line, the real-boundary completed-minus-ordinate excess is controlled exactly by $2(\cosh(\beta - \frac{1}{2}) - 1)e^{i\gamma}$. Its leading horizontal sensitivity is quadratic, matching the positive mass isolated by Fujii and later used in the spectral-zeta analysis of Bondarenko–Ivić–Saksman–Seip. In the interior upper frequency half-plane, Theorem 10.3 converts the same horizontal information into the nonnegative Laplace defect $\mathcal{D}(y)$ and gives the stated equivalence with RH. The boundary-value question at $u = 1$ remains logically separate.

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Appendix A. Numerical annex: full-data values

The centroid computation uses $N = 2{,}001{,}052$ positive ordinates, with $\gamma_N \approx 1.13249 \times 10^6$. The principal endpoint values are

$$C_N^{\zeta} = 0.776907119554635 + 0.767844590760710\,i, \tag{A.1}$$

$$C_N^{\mathrm{Lamb}} = 0.041570219277352 + 1.757541749592316\,i, \tag{A.2}$$

$$Q_N^{(H)} = 0.776905882507509 + 0.767853218705326\,i. \tag{A.3}$$

The direct and triangular reconstructions agree to 1.73×10^{-13} for the zeta centroid and 4.89×10^{-14} for the RVM centroid. The midpoint counting defect has mean 4.22×10^{-8}, standard deviation 0.3321575, and range approximately $[-1.33248, 1.29937]$.

A.1. **Tail-centroid table.**

Removed prefix	$\lvert C_\zeta \rvert$	$\lvert C_{\mathrm{RVM}} \rvert$	$\lvert C_\zeta - C_{\mathrm{RVM}} \rvert$
0	1.09230	1.75800	1.23300
10	1.59720	0.50157	1.19600
100	0.60399	0.65664	1.24920
1,000	1.34950	0.91298	1.16670
10,000	2.08820	1.20780	1.42590
100,000	2.44760	1.52210	1.01920
200,100	0.60171	1.61940	1.49940
500,260	0.92554	1.74910	1.00340
1,000,500	1.35580	1.84820	0.58972
1,500,800	2.45960	1.90640	1.41960
1,800,900	1.32320	1.93270	0.65554

A.2. **Argument-fluctuation profile.**

γ	empirical σ	Selberg scale
1,421.8	0.28310	0.31690
5,449.7	0.28880	0.33020
9,879.3	0.29921	0.33529
12,399.9	0.30184	0.33712
18,048.1	0.30601	0.34005
40,435.1	0.31372	0.34589
74,922.2	0.31907	0.35000
201,091.6	0.32669	0.35605
600,270.7	0.33476	0.36210
1,132,490.7	0.33850	0.36535

A.3. **Lambert-RVM Abel approximants.**

J	real part	imaginary part
10,000	0.041580072865611	1.757554793733856
100,000	0.041570891195329	1.757542207523970
500,000	0.041569346208409	1.757541422824472
1,000,000	0.041569178431353	1.757540990615438
2,001,052	0.041568808323635	1.757541025528465

A.4. **Imaginary-axis sensitivity scale.** For one hypothetical off-line pair with $\delta = 0.05$ at the first ordinate $\gamma_1 = 14.134725142\ldots$, the positive defect is

$$2(1 - \cos(0.05y))e^{-y\gamma_1}. \tag{A.4}$$

Its small-$y\delta$ approximation is $y^2\delta^2 e^{-y\gamma}$ and is maximized at

$$y_{\text{opt}} \approx \frac{2}{\gamma}, \tag{A.5}$$

which gives $y_{\text{opt}} \approx 0.1414955$ at γ_1. The direct pairwise values used to check the sign and scale are

y	$e^{-y\gamma_1}$	$2(1 - \cos(0.05y))e^{-y\gamma_1}$
1.0000	7.2671959×10^{-7}	1.8164205×10^{-9}
0.5000	8.5247850×10^{-4}	5.3277131×10^{-7}
0.2000	5.9193412×10^{-2}	5.9192919×10^{-6}
0.1415	1.3532668×10^{-1}	6.7738330×10^{-6}
0.1000	2.4329696×10^{-1}	6.0824114×10^{-6}
0.0500	4.9325142×10^{-1}	3.0828198×10^{-6}

Appendix B. Reproducibility files and package layout

The accompanying ZIP package contains the complete manuscript source and PDF, all figures, the CSV data used to generate them, the Python reproducibility scripts, and their console outputs. The finite zeta centroid is reported from the full dataset; the supplied Python scripts reproduce the analytic and numerical checks that do not require the complete zero table.

Appendix C. Differential identities and controlled use of the smooth-clock expansion

For numerical evaluation of (8.6), set

$$z(t) = \frac{1}{4} + \frac{it}{2}. \tag{C.1}$$

The derivatives of the exact smooth clock are then given exactly by

$$H'(t) = \frac{1}{\pi}\left[\frac{1}{2}\Re\psi(z(t)) - \frac{1}{2}\log\pi\right], \tag{C.2}$$

and

$$H''(t) = -\frac{1}{4\pi}\Im\psi_1(z(t)), \tag{C.3}$$

where ψ and ψ_1 denote the di- and trigamma functions. These identities are exact and are the preferred formulas for finite-range numerical evaluation.

The inverse-power expansion of the Riemann–Siegel theta function begins

$$\vartheta(t) \sim \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \frac{7}{5760t^3} + \frac{31}{80640t^5} + \cdots. \tag{C.4}$$

This is an asymptotic expansion, not a convergent expansion. Consequently, increasing the truncation order does not in general increase numerical accuracy. For a fixed t, the useful truncation is determined by the size of the terms and by a remainder estimate; continuing past the least term can worsen the approximation. Explicit remainder control for the log-Gamma and Riemann–Siegel theta expansions is available in [30], while refined asymptotic descriptions are discussed in [31].

There is a second, logically distinct issue. Greater accuracy in approximating the exact smooth clock H does not imply greater accuracy in approximating an individual zeta ordinate. Let τ_k be the Lambert level defined by (2.16), and let $\widehat{\tau}_k$ be the exact smooth level defined by

$$H(\widehat{\tau}_k) = k - \frac{1}{2}. \tag{C.5}$$

From (2.15) and

$$M'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}, \tag{C.6}$$

one obtains, for large k,

$$\tau_k - \widehat{\tau}_k = \frac{1}{24\tau_k\log(\tau_k/2\pi)} + O\left(\frac{1}{\tau_k^3\log\tau_k}\right). \tag{C.7}$$

Thus the first smooth-clock correction moves the Lambert level to the left for sufficiently large ordinates.

The true ordinate obeys the midpoint identity (2.6). Linearizing it about $\widehat{\tau}_k$ gives

$$\gamma_k - \widehat{\tau}_k = -\frac{S_*(\gamma_k)}{H'(\widehat{\tau}_k)} + \text{higher-order terms}. \tag{C.8}$$

The displacement of the true zero from the smooth level can therefore have either sign. A correction that makes M a more accurate approximation to H may move the corresponding smooth level toward γ_k or away from it. In particular, there is no monotone relation between the number of retained inverse-power terms and the accuracy of the resulting approximation to an individual γ_k.

For the computations associated with (8.6), equations (C.2) and (C.3) are used directly on the finite integration range. The asymptotic expansion (C.4) is used only for a tail treatment together with a controlled truncation rule and a remainder estimate. No numerical claim in this manuscript relies on the assumption that adding further inverse-power terms is automatically more accurate.

Appendix D. Python source: figure generation

```
#!/usr/bin/env python3
"""Generate the numerical figures used in the centroid manuscript.

The finite zeta tables below are transcribed from the full-data
MATLAB output for N=2,001,052 zeros. The Lambert-RVM Abel approximants are
```

```
transcribed from rvm_centroid_output.txt. The RH defect sensitivity curve is
an analytic illustration of the positive imaginary-frequency criterion.
"""
from __future__ import annotations

from pathlib import Path
import csv
import numpy as np
import matplotlib.pyplot as plt

ROOT = Path(__file__).resolve().parents[1]
FIG = ROOT / "figures"
DATA = ROOT / "data"
FIG.mkdir(parents=True, exist_ok=True)
DATA.mkdir(parents=True, exist_ok=True)

# Tail-centroid table.
removed = np.array([0, 10, 100, 1000, 10000, 100000, 200100, 500260,
          1000500, 1500800, 1800900], dtype=float)
abs_zeta = np.array([1.0923, 1.5972, 0.60399, 1.3495, 2.0882, 2.4476,
           0.60171, 0.92554, 1.3558, 2.4596, 1.3232])
abs_rvm = np.array([1.7580, 0.50157, 0.65664, 0.91298, 1.2078, 1.5221,
          1.6194, 1.7491, 1.8482, 1.9064, 1.9327])
abs_diff = np.array([1.2330, 1.1960, 1.2492, 1.1667, 1.4259, 1.0192,
           1.4994, 1.0034, 0.58972, 1.4196, 0.65554])

with (DATA / "tail_centroid_table.csv").open("w", newline="") as f:
  w = csv.writer(f)
  w.writerow(["removed_prefix", "abs_C_zeta", "abs_C_RVM", "abs_difference"])
  w.writerows(zip(removed.astype(int), abs_zeta, abs_rvm, abs_diff))

fig, ax = plt.subplots(figsize=(7.0, 4.4))
ax.plot(removed, abs_zeta, marker="o", label=r"$|C_\zeta|$")
ax.plot(removed, abs_rvm, marker="s", label=r"$|C_{\rm RVM}|$")
ax.plot(removed, abs_diff, marker="^", label=r"$|C_\zeta-C_{\rm RVM}|$")
ax.set_xscale("symlog", linthresh=10)
ax.set_xlabel("Removed prefix length")
ax.set_ylabel("Centroid modulus")
ax.set_title("Tail-centroid stability test")
ax.grid(True, alpha=0.25)
ax.legend()
fig.tight_layout()
fig.savefig(FIG / "tail_centroid_stability.pdf")
fig.savefig(FIG / "tail_centroid_stability.png", dpi=220)
plt.close(fig)

# Empirical standard-deviation profile against Selberg scale.
gamma = np.array([1421.8, 5449.7, 9879.3, 12399.9, 18048.1, 40435.1,
         74922.2, 201091.6, 600270.7, 1132490.7])
sigma_emp = np.array([0.28310, 0.28880, 0.29921, 0.30184, 0.30601,
           0.31372, 0.31907, 0.32669, 0.33476, 0.33850])
sigma_selberg = np.array([0.31690, 0.33020, 0.33529, 0.33712, 0.34005,
             0.34589, 0.35000, 0.35605, 0.36210, 0.36535])
with (DATA / "S_argument_variance_profile.csv").open("w", newline="") as f:
  w = csv.writer(f)
  w.writerow(["gamma", "sigma_empirical", "sigma_Selberg"])
  w.writerows(zip(gamma, sigma_emp, sigma_selberg))

fig, ax = plt.subplots(figsize=(7.0, 4.4))
```

```
ax.plot(gamma, sigma_emp, marker="o", label="Empirical local standard deviation")
ax.plot(gamma, sigma_selberg, marker="s", label="Selberg scale")
ax.set_xscale("log")
ax.set_xlabel(r"Zero ordinate $\gamma$")
ax.set_ylabel("Standard deviation")
ax.set_title(r"Argument fluctuation scale along the zero sequence")
ax.grid(True, alpha=0.25)
ax.legend()
fig.tight_layout()
fig.savefig(FIG / "argument_variance_profile.pdf")
fig.savefig(FIG / "argument_variance_profile.png", dpi=220)
plt.close(fig)

# Lambert-RVM Abel-series approximants.
J = np.array([10000, 100000, 500000, 1000000, 2001052], dtype=float)
re = np.array([0.041580072865611, 0.041570891195329, 0.041569346208409,
        0.041569178431353, 0.041568808323635])
im = np.array([1.757554793733856, 1.757542207523970, 1.757541422824472,
        1.757540990615438, 1.757541025528465])
ref = re[-1] + 1j * im[-1]
err = np.abs(re + 1j * im - ref)
with (DATA / "rvm_abel_approximants.csv").open("w", newline="") as f:
  w = csv.writer(f)
  w.writerow(["J", "real", "imag", "distance_to_last_approximant"])
  w.writerows(zip(J.astype(int), re, im, err))

fig, ax = plt.subplots(figsize=(7.0, 4.4))
ax.plot(J[:-1], err[:-1], marker="o")
ax.set_xscale("log")
ax.set_yscale("log")
ax.set_xlabel("Abel-series cutoff J")
ax.set_ylabel("Distance to the J=2,001,052 approximant")
ax.set_title("Lambert-RVM Abel-series convergence")
ax.grid(True, alpha=0.25)
fig.tight_layout()
fig.savefig(FIG / "rvm_abel_convergence.pdf")
fig.savefig(FIG / "rvm_abel_convergence.png", dpi=220)
plt.close(fig)

# Positive defect sensitivity on the imaginary frequency axis.
gamma1 = 14.134725142
horizontal_delta = 0.05
y = np.logspace(-2.3, 0.15, 600)
defect = 2.0 * (1.0 - np.cos(y * horizontal_delta)) * np.exp(-y * gamma1)
y_opt = 2.0 / gamma1
with (DATA / "rh_positive_defect_curve.csv").open("w", newline="") as f:
  w = csv.writer(f)
  w.writerow(["y", "pair_defect_delta_0.05_gamma1"])
  w.writerows(zip(y, defect))

fig, ax = plt.subplots(figsize=(7.0, 4.4))
ax.plot(y, defect)
ax.axvline(y_opt, linestyle="--", label=r"$2/\gamma_1$")
ax.set_xscale("log")
ax.set_yscale("log")
ax.set_xlabel(r"Imaginary frequency parameter $y$")
ax.set_ylabel("Positive paired defect")
ax.set_title(r"Sensitivity of $2(1-\cos(y\delta))e^{-y\gamma}$ for $\delta=0.05$")
ax.grid(True, alpha=0.25)
```

```
ax.legend()
fig.tight_layout()
fig.savefig(FIG / "rh_positive_defect_sensitivity.pdf")
fig.savefig(FIG / "rh_positive_defect_sensitivity.png", dpi=220)
plt.close(fig)

# Complex-plane numerical landmarks.
points = {
  "Zeta centroid": 0.776907119554635 + 0.767844590760710j,
  "Conditional analytic limit": 0.776894260793165 + 0.767855668864548j,
  "Smooth term B": 0.381782018323631 + 0.146163101528591j,
  "Arithmetic term P": 0.395112242469534 + 0.621692567335957j,
  "Lambert centroid": 0.041570219277352 + 1.757541749592316j,
}
with (DATA / "complex_landmarks.csv").open("w", newline="") as f:
  w = csv.writer(f)
  w.writerow(["label", "real", "imag"])
  for label, z in points.items():
    w.writerow([label, z.real, z.imag])

fig, ax = plt.subplots(figsize=(6.5, 5.4))
offsets = {
  "Zeta centroid": (8, 8),
  "Conditional analytic limit": (8, -10),
  "Smooth term B": (8, 8),
  "Arithmetic term P": (8, 8),
  "Lambert centroid": (8, 8),
}
for label, z in points.items():
  ax.scatter([z.real], [z.imag], s=40)
  dx, dy = offsets.get(label, (6, 6))
  ax.annotate(label, (z.real, z.imag), xytext=(dx, dy), textcoords="offset points", fontsize=8)
ax.axhline(0, linewidth=0.8)
ax.axvline(0, linewidth=0.8)
ax.set_xlabel("Real part")
ax.set_ylabel("Imaginary part")
ax.set_title("Numerical centroid landmarks")
ax.grid(True, alpha=0.25)
fig.tight_layout()
fig.savefig(FIG / "centroid_complex_landmarks.pdf")
fig.savefig(FIG / "centroid_complex_landmarks.png", dpi=220)
plt.close(fig)

print("ARTICLE FIGURE GENERATION")
print(f"output directory: {FIG}")
for p in sorted(FIG.glob("*.*")):
  if p.suffix.lower() in {".png", ".pdf"}:
    print(f"figure: {p.name}")
for p in sorted(DATA.glob("*.csv")):
  print(f"data:  {p.name}")
print(f"y_opt = 2/gamma_1 = {y_opt:.12f}")
```

Appendix E. Output: figure generation

```
ARTICLE FIGURE GENERATION
output directory: /mnt/data/zeta_centroid_article_v12/figures
figure: argument_variance_profile.pdf
figure: argument_variance_profile.png
figure: centroid_complex_landmarks.pdf
```

```
figure: centroid_complex_landmarks.png
figure: intrinsic_centered_residual_3d.pdf
figure: intrinsic_centered_residual_3d.png
figure: intro_difference_before_exponential_3d.pdf
figure: intro_difference_before_exponential_3d.png
figure: intro_paired_residual_3d.pdf
figure: intro_paired_residual_3d.png
figure: intro_zeta_vs_rvm_3d.pdf
figure: intro_zeta_vs_rvm_3d.png
figure: rh_positive_defect_sensitivity.pdf
figure: rh_positive_defect_sensitivity.png
figure: rvm_abel_convergence.pdf
figure: rvm_abel_convergence.png
figure: tail_centroid_stability.pdf
figure: tail_centroid_stability.png
data:  S_argument_variance_profile.csv
data:  complex_landmarks.csv
data:  rh_positive_defect_curve.csv
data:  rvm_abel_approximants.csv
data:  rvm_delta_summary.csv
data:  rvm_paired_residual_checkpoints.csv
data:  tail_centroid_table.csv
y_opt = 2/gamma_1 = 0.141495499906
```

Appendix F. Python source: Lambert-RVM centroid

```
#!/usr/bin/env python3
"""Reproduce the Lambert Riemann-von Mangoldt exponential centroid.

The script constructs the inverse smooth clock
  M(x)=x/(2*pi)*log(x/(2*pi*e))+7/8,
by its exact Lambert-W formula, computes the finite triangular centroid, and
checks the discrete Abel decomposition used in the analytic proof.
"""
from __future__ import annotations

import numpy as np
from scipy.special import lambertw

N = 2_001_052
pi = np.pi
k = np.arange(1, N+2, dtype=np.float64)
x = k - 11.0/8.0
tau = 2*pi*x / lambertw(x/np.e).real
z = np.exp(1j*tau)

# Direct triangular centroid for k=1,...,N.
w = (N - np.arange(1, N+1) + 1)/N
C_direct = np.sum(w*z[:N], dtype=np.complex128)

# Gap/Abel coefficients.
gap = np.diff(tau)            # g_k=tau_{k+1}-tau_k
d = 1.0/(np.exp(1j*gap)-1.0)      # d_k
Dd = np.diff(d)             # d_{k+1}-d_k

first_term = -d[0]*z[0]
raw_series_trunc = first_term - np.sum(Dd*z[1:N], dtype=np.complex128)
boundary_average = np.sum(d[:N]*z[1:N+1], dtype=np.complex128)/N
weighted_series = -np.sum((1.0-np.arange(1,N)/N)*Dd*z[1:N], dtype=np.complex128)
```

```
C_abel = first_term + boundary_average + weighted_series

# Verify the inverse clock numerically.
M_tau = tau[:N]/(2*pi)*np.log(tau[:N]/(2*pi*np.e)) + 7.0/8.0
clock_err = M_tau - (np.arange(1,N+1)-0.5)

# Coarse tail diagnostics for the convergent Abel series.
cutoffs = np.array([10_000, 100_000, 500_000, 1_000_000, N], dtype=int)
partials = []
for J in cutoffs:
  ps = first_term - np.sum(Dd[:J-1]*z[1:J], dtype=np.complex128)
  partials.append(ps)

print('LAMBERT RVM EXPONENTIAL CENTROID REPRODUCTION')
print(f'N              = {N}')
print(f'tau_1            = {tau[0]:.15f}')
print(f'tau_N            = {tau[N-1]:.15f}')
print(f'max |M(tau_k)-(k-1/2)|    = {np.max(np.abs(clock_err)):.12e}')
print('')
print(f'C_N direct          = {C_direct.real:+.15f} {C_direct.imag:+.15f}i')
print(f'-d_1 exp(i tau_1)      = {first_term.real:+.15f} {first_term.imag:+.15f}i')
print(f'boundary average       = {boundary_average.real:+.15e} {boundary_average.imag:+.15e}i')
print(f'weighted Abel series     = {weighted_series.real:+.15f} {weighted_series.imag:+.15f}i')
print(f'C_N from Abel identity    = {C_abel.real:+.15f} {C_abel.imag:+.15f}i')
print(f'|direct-Abel|        = {abs(C_direct-C_abel):.12e}')
print('')
print(f'infinite-series truncation N = {raw_series_trunc.real:+.15f} {raw_series_trunc.imag:+.15f}i')
print('partial Abel-series approximants:')
for J, ps in zip(cutoffs, partials):
  print(f' J={J:8d}: {ps.real:+.15f} {ps.imag:+.15f}i')
print('')
print('The analytic theorem in the manuscript proves convergence of the')
print('unweighted Abel series and an O(log J / J) tail at the smooth-clock scale.')
```

Appendix G. Output: Lambert-RVM centroid

```
LAMBERT RVM EXPONENTIAL CENTROID REPRODUCTION
N              = 2001052
tau_1            = 14.521346953065626
tau_N            = 1132490.722587621537969
max |M(tau_k)-(k-1/2)|    = 6.984919309616e-10

C_N direct          = +0.041570227040394 +1.757541761467938i
-d_1 exp(i tau_1)      = +6.031991313792247 +2.977851908778479i
boundary average       = -1.700645406387667e-06 +1.385344506745871e-07i
weighted Abel series     = -5.990419386106493 -1.220310285845148i
C_N from Abel identity    = +0.041570227040348 +1.757541761467782i
|direct-Abel|        = 1.625736540272e-13

infinite-series truncation N = +0.041568808323635 +1.757541025528465i
partial Abel-series approximants:
 J=   10000: +0.041580072865611 +1.757554793733856i
 J=  100000: +0.041570891195329 +1.757542207523970i
 J=  500000: +0.041569346208409 +1.757541422824472i
 J= 1000000: +0.041569178431353 +1.757540990615438i
 J= 2001052: +0.041568808323635 +1.757541025528465i

The analytic theorem in the manuscript proves convergence of the
unweighted Abel series and an O(log J / J) tail at the smooth-clock scale.
```

Appendix H. Python source: conditional zeta constant

```
#!/usr/bin/env python3
"""Reproduce the conditional one-sided analytic centroid constant.

This script evaluates the explicit quantities B(x0) and P(x0) used in the
manuscript at x0=1 and sigma=1/2. It does not use a table of zeta zeros.

The mathematical identification of B+P with the ordinal triangular centroid
is a theorem only under the manuscript's stated boundary-interchange/Riesz
compatibility hypothesis (and RH for the pure ordinate phase).
"""
from __future__ import annotations

import mpmath as mp

mp.mp.dps = 70

x0 = mp.mpf(1)
sigma = mp.mpf('0.5')
U = mp.mpf(20)


def theta(t: mp.mpf) -> mp.mpf:
  z = mp.mpf('0.25') + 0.5j*t
  # Continuous branch near positive t; for x0=1 principal loggamma is valid.
  return mp.im(mp.loggamma(z)) - mp.mpf('0.5')*t*mp.log(mp.pi)


def H(t: mp.mpf) -> mp.mpf:
  return 1 + theta(t)/mp.pi


def Hp(t: mp.mpf) -> mp.mpf:
  z = mp.mpf('0.25') + 0.5j*t
  return (mp.mpf('0.5')*mp.re(mp.digamma(z)) - mp.mpf('0.5')*mp.log(mp.pi))/mp.pi


def Hpp(t: mp.mpf) -> mp.mpf:
  z = mp.mpf('0.25') + 0.5j*t
  return -mp.im(mp.polygamma(1, z))/(4*mp.pi)


def continuous_log_zeta(s: mp.mpc) -> mp.mpc:
  # At Im(s)=+/-1 and Re(s)>=1/2 the horizontal ray contains no zeta zero
  # or pole; the principal value agrees with continuation from Re(s)>>1
  # for this numerical evaluation.
  return mp.log(mp.zeta(s))


def primes_upto(n: int) -> list[int]:
  sieve = bytearray(b'\x01')*(n+1)
  sieve[0:2] = b'\x00\x00'
  p = 2
  while p*p <= n:
    if sieve[p]:
      sieve[p*p:n+1:p] = b'\x00' * (((n-p*p)//p)+1)
    p += 1
  return [i for i in range(2,n+1) if sieve[i]]
```

```
def euler_product_tail_i2(U: mp.mpf, prime_max: int = 1000, m_max: int = 30) -> mp.mpc:
  """Tail U..infinity of e^u[Log zeta(sigma+u+ix0)-2^{-s}]du.

  For u>=U use the absolutely convergent Euler expansion
   Log zeta(s) = sum_p sum_{m>=1} p^{-ms}/m.
  The (p,m)=(2,1) term is subtracted in the integrand.
  """
  total = mp.mpc(0)
  for p in primes_upto(prime_max):
    lp = mp.log(p)
    for m in range(1, m_max+1):
      if p == 2 and m == 1:
        continue
      rate = m*lp - 1
      coeff = (mp.power(p, -m*(sigma + 1j*x0)) / m)
      term = coeff * mp.e**(-rate*U) / rate
      total += term
      if abs(term) < mp.mpf('1e-75'):
        break
  return total


# Smooth (Gamma/Riemann-Siegel) contribution.
# We integrate H'' exactly on [x0,x_cut] and use the standard asymptotic
# expansion of theta(t) for the oscillatory tail. This avoids a very slow
# infinite-interval quadrature while keeping the computation transparent.
x_cut = mp.mpf(30)
finite_osc = mp.quad(lambda t: mp.e**(1j*t)*Hpp(t), [x0, 2, 4, 8, 16, x_cut])


def oscillatory_power_tail(p: int, x: mp.mpf, extra_steps: int = 45) -> mp.mpc:
  """J_p(x)=int_x^inf exp(i t)t^{-p}dt by downward recurrence."""
  # Repeated integration by parts gives
  # J_p = i e^{ix} x^{-p} - i p J_{p+1}.
  top = p + extra_steps
  J = mp.mpc(0)
  for q in range(top-1, p-1, -1):
    J = 1j*mp.e**(1j*x)*x**(-q) - 1j*q*J
  return J


tail_osc = oscillatory_power_tail(1, x_cut)/(2*mp.pi)
# theta(t) = main + sum_{n>=1} c_n t^{-(2n-1)}.
# Hence H'' adds c_n(2n-1)(2n)/(pi t^{2n+1}).
for n in range(1, 13):
  c_n = ((-1)**n * mp.bernpoly(2*n, mp.mpf('0.5'))
      / (4*n*(2*n-1)))
  p = 2*n + 1
  coeff = c_n*(2*n-1)*(2*n)/mp.pi
  tail_osc += coeff*oscillatory_power_tail(p, x_cut)

osc = finite_osc + tail_osc
B = mp.e**(1j*x0)*(H(x0) + 1j*Hp(x0)) + 1j*osc

# Arithmetic one-sided contribution.
I1 = mp.quad(lambda u: mp.e**(-u)*continuous_log_zeta(sigma+u-1j*x0), [0, 1, 2, 4, 8, 16, 32, 64])
I2_head = mp.quad(
```

```
  lambda u: mp.e**u*(continuous_log_zeta(sigma+u+1j*x0) - mp.power(2, -(sigma+u+1j*x0))),
  [0, 1, 2, 4, 8, 12, 16, U]
)
I2_tail = euler_product_tail_i2(U)
I2 = I2_head + I2_tail
sub = mp.power(2, -(sigma+1j*x0))/(1-mp.log(2))
P = 1j*mp.e**(1j*x0)/(2*mp.pi)*(I1 + I2 - sub)
Cstar = B + P
real_closed = (mp.e - 1/(mp.e**2-1))/(2*mp.sqrt(mp.e))

fmt = lambda z: mp.nstr(z, 50)
print('ZETA CONDITIONAL CENTROID CONSTANT REPRODUCTION')
print('x0         =', fmt(x0))
print('sigma        =', fmt(sigma))
print('Euler-tail split U =', fmt(U))
print('theta-tail x_cut   =', fmt(x_cut))
print('')
print('B(x0)        =', fmt(B))
print('I1          =', fmt(I1))
print('I2 head       =', fmt(I2_head))
print('I2 Euler tail    =', fmt(I2_tail))
print('I2 total       =', fmt(I2))
print('P(x0)        =', fmt(P))
print('C_infty_zeta        =', fmt(Cstar))
print('closed Re(C_infty_zeta)  =', fmt(real_closed))
print('Re difference    =', fmt(mp.re(Cstar)-real_closed))
print('')
print('C_infty_zeta Re      =', mp.nstr(mp.re(Cstar), 30))
print('C_infty_zeta Im      =', mp.nstr(mp.im(Cstar), 30))
```

Appendix I. Output: conditional zeta constant

```
ZETA CONDITIONAL CENTROID CONSTANT REPRODUCTION
x0         = 1.0
sigma        = 0.5
Euler-tail split U = 20.0
theta-tail x_cut   = 30.0

B(x0)        = (0.381782018323631154326709791323034084652963196262 33 +
     0.146163101528591363602053435096049848385971434683 62j)
I1          = (0.061165767995974675659964658117807665570719077369662 +
     0.788133849068108241703167414010045356582697140906 94j)
I2 head       = (1.362402112666911852152298732408424069505404596561 8 -
     6.163046818757005673058295466119728280288804144959 9j)
I2 Euler tail    = (0.370583027258264382323026140260004718198422231656 52 -
     0.725796076066282284190762190421209202275133183033 9j)
I2 total       = (1.732985139925176234475324872668428787703826828218 3 -
     6.888842894823287957249057656540937482563937327993 8j)
P(x0)        = (0.395112242469534050693365268458680515507561878817 91 +
     0.621692567335957076475858204211908808296261392491 38j)
C_infty_zeta        = (0.776894260793165205020075059781714600160525075080 24 +
     0.767855668864548440077911639307958656682232827174 99j)
closed Re(C_infty_zeta)  = 0.77689426079325358170733757683887017228268849806797
Re difference    = -0.000000000000088376687262517057155572122163422987724506269883362

C_infty_zeta Re      = 0.77689426079316520502007505978
C_infty_zeta Im      = 0.76785566886454844007791163930
```

Appendix J. Python source: centroid-family RH equivalence

```
#!/usr/bin/env python3
"""Numerical checks for the damped centroid RH-equivalence section.

This script verifies the finite algebra used in the positive Laplace
defect and reproduces the sensitivity table for one hypothetical
off-critical pair.
"""
from __future__ import annotations

import math

GAMMA1 = 14.1347251417
DELTA = 0.05
Y_VALUES = [1.0, 0.5, 0.2, 0.1415, 0.1, 0.05]


def pair_defect(y: float, delta: float = DELTA, gamma: float = GAMMA1) -> float:
  """P_pair(y)=2(1-cos(y*delta))*exp(-y*gamma)."""
  return 2.0 * (1.0 - math.cos(y * delta)) * math.exp(-y * gamma)


def complete_pair(y: float, delta: float = DELTA, gamma: float = GAMMA1) -> complex:
  """Completed-zero pair at complex frequency omega=i y."""
  return (
    complex(math.cos(y * delta), math.sin(y * delta))
    + complex(math.cos(-y * delta), math.sin(-y * delta))
  ) * math.exp(-y * gamma)


def ordinate_pair(y: float, gamma: float = GAMMA1) -> float:
  """Ordinate-only pair at complex frequency omega=i y."""
  return 2.0 * math.exp(-y * gamma)


print("DAMPED CENTROID RH-EQUIVALENCE NUMERICAL CHECK")
print(f"gamma_1             = {GAMMA1:.13f}")
print(f"hypothetical delta       = {DELTA:.8f}")
print(f"universal logical range     = 0 < y <= {4*math.pi:.15f}")
print("")
print(" y    exp(-y gamma_1)    positive pair defect    direct difference")
for y in Y_VALUES:
  damp = math.exp(-y * GAMMA1)
  p = pair_defect(y)
  direct = ordinate_pair(y) - complete_pair(y).real
  print(f"{y:6.4f}  {damp: .12e}    {p: .12e}    {direct: .12e}")

print("")
y_opt = 2.0 / GAMMA1
print(f"quadratic-model optimum 2/gamma_1 = {y_opt:.15f}")
print(f"pair defect at y_opt       = {pair_defect(y_opt):.12e}")
print("")
# Universal injectivity check underlying the theorem: if |delta|<1/2 and
# 0<y<=4*pi, then |y*delta|<2*pi. Hence cos(y*delta)=1 forces delta=0.
# We print the endpoint bound to make the strict inequality explicit.
print("At y=4*pi and |delta|<1/2, one has |y*delta|<2*pi;")
print("therefore the only possible multiple of 2*pi is zero.")
```

Appendix K. Output: centroid-family RH equivalence

```
DAMPED CENTROID RH-EQUIVALENCE NUMERICAL CHECK
gamma_1             = 14.1347251417000
hypothetical delta       = 0.05000000
universal logical range     = 0 < y <= 12.566370614359172

 y    exp(-y gamma_1)    positive pair defect    direct difference
1.0000   7.267195903693e-07     1.816420507677e-09     1.816420507677e-09
0.5000   8.524784984792e-04     5.327713121764e-07     5.327713121764e-07
0.2000   5.919341192473e-02     5.919291864796e-06     5.919291864789e-06
0.1415   1.353266751645e-01     6.773833048767e-06     6.773833048757e-06
0.1000   2.432969624240e-01     6.082411388894e-06     6.082411388886e-06
0.0500   4.932514190795e-01     3.082819763621e-06     3.082819763578e-06

quadratic-model optimum 2/gamma_1 = 0.141495499908918
pair defect at y_opt       = 6.773833057413e-06

At y=4*pi and |delta|<1/2, one has |y*delta|<2*pi;
therefore the only possible multiple of 2*pi is zero.
```

Appendix L. Python source: zeta–RVM residual walks

```
#!/usr/bin/env python3
"""Reproduce the zeta/Lambert exponential walks and the residual constructions.

Input files are local binary arrays of IEEE-754 float64 values:
 - zeross.bin: positive zeta-zero ordinates, from Odlyzko's public table.
 - zLamb.bin: paired Lambert levels.

The mathematical objects are
 Z_k = exp(i gamma_k),
 V_k = exp(i tau_k),
 b_k = Z_k - V_k,
 u_k = exp(i (gamma_k-tau_k)),
 a_k = u_k - 1.

The script writes figures in PNG/PDF, a complete text report, and checkpoint CSVs.
"""
from __future__ import annotations

import argparse
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.gridspec import GridSpec


def triangular_centroid(x: np.ndarray) -> complex:
  n = x.size
  w = np.arange(n, 0, -1, dtype=np.float64) / n
  return np.sum(w * x)


def cfmt(z: complex) -> str:
  return f"{z.real:.15f} {z.imag:+.15f}i"


def select_indices(n: int, target: int = 6000) -> np.ndarray:
  if n <= target:
    return np.arange(n)
```

```
  return np.unique(np.linspace(0, n - 1, target, dtype=np.int64))


def save_figure(fig: plt.Figure, stem: Path) -> None:
  fig.savefig(stem.with_suffix('.png'), dpi=220, bbox_inches='tight', pad_inches=0.25)
  fig.savefig(stem.with_suffix('.pdf'), bbox_inches='tight', pad_inches=0.25)
  plt.close(fig)


def complex_limits(*arrays: np.ndarray, pad: float = 0.04):
  re_min = min(np.min(np.real(a)) for a in arrays)
  re_max = max(np.max(np.real(a)) for a in arrays)
  im_min = min(np.min(np.imag(a)) for a in arrays)
  im_max = max(np.max(np.imag(a)) for a in arrays)
  cx = 0.5 * (re_min + re_max)
  cy = 0.5 * (im_min + im_max)
  radius = 0.5 * max(re_max - re_min, im_max - im_min)
  radius *= (1.0 + pad)
  if radius == 0:
    radius = 1.0
  return (cx - radius, cx + radius), (cy - radius, cy + radius)


def make_dual_walk_figure(stem: Path, z: np.ndarray, curve1: np.ndarray, curve2: np.ndarray,
             label1: str, label2: str, title3d: str, title2d: str):
  idx = select_indices(curve1.size, target=5000)
  xlim, ylim = complex_limits(curve1[idx], curve2[idx], pad=0.06)
  fig = plt.figure(figsize=(12.6, 5.6))
  gs = GridSpec(1, 2, figure=fig, width_ratios=[1.45, 1.0])
  ax3 = fig.add_subplot(gs[0, 0], projection='3d')
  ax3.plot(curve1.real[idx], curve1.imag[idx], z[idx], linewidth=0.30, label=label1)
  ax3.plot(curve2.real[idx], curve2.imag[idx], z[idx], linewidth=0.30, label=label2)
  ax3.set_xlim(*xlim)
  ax3.set_ylim(*ylim)
  ax3.set_xlabel('Real part')
  ax3.set_ylabel('Imaginary part')
  ax3.set_zlabel(r'$k$', labelpad=-2)
  ax3.set_title(title3d)
  ax3.legend(loc='upper left', fontsize=8)
  ax2 = fig.add_subplot(gs[0, 1])
  ax2.plot(curve1.real[idx], curve1.imag[idx], linewidth=0.28, label=label1)
  ax2.plot(curve2.real[idx], curve2.imag[idx], linewidth=0.28, label=label2)
  ax2.set_xlim(*xlim)
  ax2.set_ylim(*ylim)
  ax2.set_aspect('equal', adjustable='box')
  ax2.axhline(0.0, linewidth=0.6)
  ax2.axvline(0.0, linewidth=0.6)
  ax2.set_xlabel('Real part')
  ax2.set_ylabel('Imaginary part')
  ax2.set_title(title2d)
  ax2.legend(loc='best', fontsize=8)
  fig.tight_layout()
  save_figure(fig, stem)


def make_single_walk_figure(stem: Path, z: np.ndarray, curve: np.ndarray, title3d: str, title2d: str,
              xlabel3d: str, ylabel3d: str):
  idx = select_indices(curve.size, target=7000)
  xlim, ylim = complex_limits(curve[idx], pad=0.06)
```

```
  fig = plt.figure(figsize=(12.0, 5.5))
  gs = GridSpec(1, 2, figure=fig, width_ratios=[1.45, 1.0])
  ax3 = fig.add_subplot(gs[0, 0], projection='3d')
  ax3.plot(curve.real[idx], curve.imag[idx], z[idx], linewidth=0.30)
  ax3.set_xlim(*xlim)
  ax3.set_ylim(*ylim)
  ax3.set_xlabel(xlabel3d)
  ax3.set_ylabel(ylabel3d)
  ax3.set_zlabel(r'$k$', labelpad=-2)
  ax3.set_title(title3d)
  ax2 = fig.add_subplot(gs[0, 1])
  ax2.plot(curve.real[idx], curve.imag[idx], linewidth=0.28)
  ax2.set_xlim(*xlim)
  ax2.set_ylim(*ylim)
  ax2.set_aspect('equal', adjustable='box')
  ax2.axhline(0.0, linewidth=0.6)
  ax2.axvline(0.0, linewidth=0.6)
  ax2.set_xlabel('Real part')
  ax2.set_ylabel('Imaginary part')
  ax2.set_title(title2d)
  fig.tight_layout()
  save_figure(fig, stem)


def main() -> None:
  parser = argparse.ArgumentParser()
  parser.add_argument('--zeros', type=Path, default=Path('zeross.bin'))
  parser.add_argument('--rvm', type=Path, default=Path('zLamb.bin'))
  parser.add_argument('--outdir', type=Path, default=Path('.'))
  args = parser.parse_args()

  outdir = args.outdir.resolve()
  figdir = outdir / 'figures'
  datadir = outdir / 'data'
  outputdir = outdir / 'outputs'
  figdir.mkdir(parents=True, exist_ok=True)
  datadir.mkdir(parents=True, exist_ok=True)
  outputdir.mkdir(parents=True, exist_ok=True)

  gamma = np.fromfile(args.zeros, dtype=np.float64)
  tau = np.fromfile(args.rvm, dtype=np.float64)
  if gamma.size != tau.size:
    raise ValueError(f'Length mismatch: {gamma.size} zeta ordinates vs {tau.size} Lambert levels')
  if gamma.size == 0:
    raise ValueError('Input arrays are empty')

  n_total = gamma.size
  n_axis = np.arange(1, n_total + 1, dtype=np.float64)
  delta = gamma - tau
  zeta_phase = np.exp(1j * gamma)
  lamb_phase = np.exp(1j * tau)
  paired_residual = zeta_phase - lamb_phase
  intrinsic_phase = np.exp(1j * delta)
  intrinsic_residual = intrinsic_phase - 1.0

  s_zeta = np.cumsum(zeta_phase)
  s_lamb = np.cumsum(lamb_phase)
  r_pair = np.cumsum(paired_residual)
  s_delta = np.cumsum(intrinsic_phase)
```

```
r_intrinsic = np.cumsum(intrinsic_residual)

c_zeta = triangular_centroid(zeta_phase)
c_lamb = triangular_centroid(lamb_phase)
c_pair = triangular_centroid(paired_residual)
c_delta = triangular_centroid(intrinsic_phase)
c_intrinsic = triangular_centroid(intrinsic_residual)

c_pair_identity_error = abs(c_pair - (c_zeta - c_lamb))
intrinsic_shift_error = abs(c_delta - (c_intrinsic + (n_total + 1) / 2.0))
b_mod_identity_error = np.max(np.abs(np.abs(paired_residual) - 2.0 * np.abs(np.sin(delta / 2.0))))

requested = np.array([10**3, 10**4, 10**5, 5*10**5, 10**6, n_total], dtype=np.int64)
checkpoints = np.unique(requested[requested <= n_total])
rows = []
abs_r = np.abs(r_pair)
running_max = np.maximum.accumulate(abs_r)
for m in checkpoints:
  rows.append((m, abs_r[m-1], running_max[m-1]))
np.savetxt(
  datadir / 'rvm_paired_residual_checkpoints.csv',
  np.asarray(rows),
  delimiter=',',
  header='N,abs_R_N,max_abs_R_k_up_to_N',
  comments='',
  fmt=['%d', '%.15e', '%.15e'],
)

delta_stats = np.array([[np.mean(delta), np.std(delta), np.min(delta), np.max(delta), np.mean(delta > 0)]])
np.savetxt(
  datadir / 'rvm_delta_summary.csv',
  delta_stats,
  delimiter=',',
  header='mean_delta,std_delta,min_delta,max_delta,share_positive',
  comments='',
  fmt='%.15e',
)

n_intro = min(100_000, n_total)
z_intro = np.arange(1, n_intro + 1, dtype=np.float64)
make_dual_walk_figure(
  figdir / 'intro_zeta_vs_rvm_3d',
  z_intro,
  s_zeta[:n_intro],
  s_lamb[:n_intro],
  'True-zero walk',
  'Lambert walk',
  'Three-dimensional ordered exponential walks',
  'Projection on the complex plane',
)

make_single_walk_figure(
  figdir / 'intro_paired_residual_3d',
  n_axis,
  r_pair,
  r'Three-dimensional paired residual $R_k$',
  r'Complex-plane projection of $R_k$',
  r'$\Re R_k$',
  r'$\Im R_k$'
```

```
)

make_single_walk_figure(
  figdir / 'intro_difference_before_exponential_3d',
  n_axis,
  s_delta,
  r'Three-dimensional walk $U_k=\sum_{j\leq k} e^{i(\gamma_j-\tau_j)}$',
  r'Complex-plane projection of $U_k$',
  r'$\Re U_k$',
  r'$\Im U_k$'
)

make_single_walk_figure(
  figdir / 'intrinsic_centered_residual_3d',
  n_axis,
  r_intrinsic,
  r'Three-dimensional centered intrinsic residual $A_k$',
  r'Complex-plane projection of $A_k$',
  r'$\Re A_k$',
  r'$\Im A_k$'
)

energy_scale = float(np.sqrt(np.sum(np.abs(paired_residual)**2)))
final_pair_abs = float(abs(r_pair[-1]))
max_pair_abs = float(np.max(np.abs(r_pair)))
max_pair_idx = int(np.argmax(np.abs(r_pair)) + 1)

report = []
report.append('RVM/ZETA EXPONENTIAL RESIDUAL AUDIT')
report.append('====================================')
report.append(f'zeros_file = {args.zeros.resolve()}')
report.append(f'lambert_file = {args.rvm.resolve()}')
report.append(f'N = {n_total}')
report.append(f'gamma_1 = {gamma[0]:.15f}')
report.append(f'gamma_N = {gamma[-1]:.15f}')
report.append(f'tau_1 = {tau[0]:.15f}')
report.append(f'tau_N = {tau[-1]:.15f}')
report.append('')
report.append('DELTA STATISTICS')
report.append('----------------')
report.append(f'mean(delta) = {np.mean(delta):.15e}')
report.append(f'std(delta) = {np.std(delta):.15e}')
report.append(f'min(delta) = {np.min(delta):.15e}')
report.append(f'max(delta) = {np.max(delta):.15e}')
report.append(f'share(delta>0) = {np.mean(delta > 0):.15e}')
report.append('')
report.append('FINAL PARTIAL SUMS')
report.append('------------------')
report.append(f'S_zeta(N) = {cfmt(s_zeta[-1])}')
report.append(f'S_Lambert(N) = {cfmt(s_lamb[-1])}')
report.append(f'R_pair(N) = {cfmt(r_pair[-1])}')
report.append(f'|R_pair(N)| = {final_pair_abs:.15e}')
report.append(f'max_n |R_pair(n)| = {max_pair_abs:.15e}')
report.append(f'argmax_n |R_pair(n)| = {max_pair_idx}')
report.append(f'sqrt(sum |b_k|^2) = {energy_scale:.15e}')
report.append(f'|R_pair(N)| / sqrt(sum |b_k|^2) = {final_pair_abs/energy_scale:.15e}')
report.append(f'S_delta(N) = {cfmt(s_delta[-1])}')
report.append(f'A_intrinsic(N) = {cfmt(r_intrinsic[-1])}')
report.append('')
```

```
  report.append('TRIANGULAR CENTROIDS')
  report.append('--------------------')
  report.append(f'C_zeta = {cfmt(c_zeta)}')
  report.append(f'C_Lambert = {cfmt(c_lamb)}')
  report.append(f'C_pair = {cfmt(c_pair)}')
  report.append(f'C_zeta - C_Lambert = {cfmt(c_zeta-c_lamb)}')
  report.append(f'identity error C_pair-(C_zeta-C_Lambert) = {c_pair_identity_error:.15e}')
  report.append(f'C_delta_phase = {cfmt(c_delta)}')
  report.append(f'C_intrinsic_residual = {cfmt(c_intrinsic)}')
  report.append(f'identity error C_delta-[C_intrinsic+(N+1)/2] = {intrinsic_shift_error:.15e}')
  report.append('')
  report.append('TERM-BY-TERM IDENTITY CHECK')
  report.append('---------------------------')
  report.append(f'max ||b_k|-2|sin(delta_k/2)|| = {b_mod_identity_error:.15e}')
  report.append('')
  report.append('PAIRED-RESIDUAL CHECKPOINTS')
  report.append('---------------------------')
  report.append('N, |R_N|, max_{k<=N}|R_k|')
  for m, val, mx in rows:
    report.append(f'{m}, {val:.15e}, {mx:.15e}')
  report.append('')
  report.append('FIGURES WRITTEN')
  report.append('---------------')
  for stem in [
    'intro_zeta_vs_rvm_3d',
    'intro_paired_residual_3d',
    'intro_difference_before_exponential_3d',
    'intrinsic_centered_residual_3d',
  ]:
    report.append(str((figdir / (stem + '.png')).resolve()))
    report.append(str((figdir / (stem + '.pdf')).resolve()))

  report_text = '\n'.join(report) + '\n'
  report_file = outputdir / 'rvm_residual_walks_output.txt'
  report_file.write_text(report_text, encoding='utf-8')
  print(report_text, end='')


if __name__ == '__main__':
  main()
```

Appendix M. Output: zeta–RVM residual walks

```
RVM/ZETA EXPONENTIAL RESIDUAL AUDIT
===================================
zeros_file = /mnt/data/zeta_centroid_article_v14/zeross.bin
lambert_file = /mnt/data/zeta_centroid_article_v14/zLamb.bin
N = 2001052
gamma_1 = 14.134725142000001
gamma_N = 1132490.658714411081746
tau_1 = 14.521346953065626
tau_N = 1132490.722587621537969

DELTA STATISTICS
----------------
mean(delta) = -1.730666072268319e-07
std(delta) = 1.880662997769061e-01
min(delta) = -9.478608912141624e-01
max(delta) = 9.937108509875827e-01
```

```
share(delta>0) = 5.001574171985536e-01

FINAL PARTIAL SUMS
------------------
S_zeta(N) = 0.357937659134386 +1.374928508944702i
S_Lambert(N) = -0.416978017681934 +3.650705195686669i
R_pair(N) = 0.774915676816428 -2.275776686742230i
|R_pair(N)| = 2.404091061939066e+00
max_n |R_pair(n)| = 3.006826277026504e+00
argmax_n |R_pair(n)| = 1079889
sqrt(sum |b_k|^2) = 2.649282445682294e+02
|R_pair(N)| / sqrt(sum |b_k|^2) = 9.074498892547935e-03
S_delta(N) = 1965958.512614915147424 -0.311563413281028i
A_intrinsic(N) = -35093.487385001448274 -0.311563413281028i

TRIANGULAR CENTROIDS
--------------------
C_zeta = 0.776907119554533 +0.767844590760585i
C_Lambert = 0.041570219277355 +1.757541749592341i
C_pair = 0.735336900277166 -0.989697158831749i
C_zeta - C_Lambert = 0.735336900277178 -0.989697158831756i
identity error C_pair-(C_zeta-C_Lambert) = 1.393761091115239e-14
C_delta_phase = 981742.505021953489631 -0.175254737451056i
C_intrinsic_residual = -18783.994978046514007 -0.175254737451056i
identity error C_delta-[C_intrinsic+(N+1)/2] = 0.000000000000000e+00

TERM-BY-TERM IDENTITY CHECK
---------------------------
max ||b_k|-2|sin(delta_k/2)|| = 2.220446049250313e-16

PAIRED-RESIDUAL CHECKPOINTS
---------------------------
N, |R_N|, max_{k<=N}|R_k|
1000, 9.947318248506007e-01, 2.818441440301175e+00
10000, 2.292366849014799e+00, 2.843940611147940e+00
100000, 2.198062492064551e+00, 2.896392475908556e+00
500000, 1.691352787832004e+00, 2.969560403811724e+00
1000000, 2.427442430097463e+00, 2.986930002751869e+00
2001052, 2.404091061939066e+00, 3.006826277026504e+00

FIGURES WRITTEN
---------------
/mnt/data/zeta_centroid_article_v14/figures/intro_zeta_vs_rvm_3d.png
/mnt/data/zeta_centroid_article_v14/figures/intro_zeta_vs_rvm_3d.pdf
/mnt/data/zeta_centroid_article_v14/figures/intro_paired_residual_3d.png
/mnt/data/zeta_centroid_article_v14/figures/intro_paired_residual_3d.pdf
/mnt/data/zeta_centroid_article_v14/figures/intro_difference_before_exponential_3d.png
/mnt/data/zeta_centroid_article_v14/figures/intro_difference_before_exponential_3d.pdf
/mnt/data/zeta_centroid_article_v14/figures/intrinsic_centered_residual_3d.png
/mnt/data/zeta_centroid_article_v14/figures/intrinsic_centered_residual_3d.pdf
```